Mount Kenya University

Module 1: Certificate in fundamentals of Competency Based Education

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1.0 Preamble

The Competency-Based Curriculum (CBC) is conceptualized to be implemented using a structure that seeks to enable every Kenyan learner to become an **Engaged, Empowered and Ethical Citizen**. In addition, this structure must align the learner to the world-class standards in skills and knowledge required in the 21st Century**. Since 2017**, Kenya has been implementing the 2:6:3:3 structure in basic education, organised into **age-based** and **stage-based** progression pathways. The age-based pathway is organised into three (3) levels: Early Years Education, Middle School Education and Senior School. The Stage-based pathway for SNE is organised into four levels, namely: Foundational, Intermediate, Pre-vocational and Vocational. Currently, the curriculum has been rolled-out up to grade 8. **The school is required to train teachers who can handle learners from pre-primary 1 to grade 12** and **in both pathways**.

The **Presidential Working Party on Education Reforms (PWPER) taskforce report** was presented to the President **on 1st August 2023.** The recommendations for the Teacher Education are detailed in section **4.5.2 (pg 71)** of the report. Key among these are:

1. 4.5.2.1 …*The teachers who have graduated before 2023 undergo a mandatory one-year retooling program for compliance with the curriculum change*

2. 4.5.2.12- *Ensure that the current students pursuing Bachelor of Education programme graduation from 2023 onwards are adequately trained in CBE. All universities offering (TE-Teacher education) should ensure their programmes are aligned with CBTE (Competency Based Teacher Education) and CBTA (Competency Based Teacher Assessment). The minimum duration for a Bachelor of Education degree in Teacher Education is four years (3 years for professional courses and 1 year for practicum).*

2.0 Actions taken by the School of Education

1. The School of Education has aligned the **six Bachelor of Education Programmes** and one postgraduate Diploma in Education (PGDE) programme to **CBC** and to CUE guidelines for Teacher Education Programmes (TEP) December 2023 guidelines. The school has already submitted the programmes to CUE for accreditation. **The Senate approved the Curriculum on 29th February, 2024.** The new curriculum will be operational from **September 2024**.

2. The school developed a Proficiency Certificate Course in **Fundamentals of Competency-Based Education** to be undertaken by *have* ***graduated before 2023***

3.0 School Recommendations

a) For the Continuing students:

1) One course unit-**BEM1108: Fundamentals of Competency-Based Education** to be included in the current SIX (6) B.Ed Curricula namely BEDA, BEDS, BEPO, BSNES, BSNEP and BECD.
2) One Course Unit-**GEM 5102: Fundamentals of Competency-Based Education** to be included in the current Post Graduate Diploma Education Curriculum
3) The University offer the **Fundamentals of Competency-Based Education** course unit to **8 B.Ed cohorts** starting with **Jan-April 2022 intake** and ending with **May-August 2024** intake and **3 PGDE** cohorts starting with **April 2023**
4) The unit to be included in the **students' transcript**. This will increase their employability and also marketability of the B.Ed and PGDE Courses in the University.
5) The University to **absorb the cost of teaching** the extra **ONE COURSE UNIT** as it may not be tenable to charge the students extra fees.
6) From **September, 2024, new fees will apply for the revised curriculum.**

b) **For Alumni**:

1) To be offered Proficiency Certificate Course in **Fundamentals of Competency-Based Education** using Blended Learning during school holidays.
2) The Certificate Course to be domiciled in the CPD
3) The Certificate be costed at the rate of other certificate courses at the centre (Ksh 10,000)

4. Our Prayer:

- Senate consider the recommendations for immediate action by the School.

Appendix 1: Implementation of teaching- BEM1108: Fundamentals of Competency-Based Education

Week	Strands/Sub-strands	Duration (hrs)
1	Basic Educational Framework for CBC	3
2 and 3	**Interpretations of Curriculum Design-** National Goals of Education, level of learning outcomes, learning area general outcomes, Essence statement, Strands and Sub strands Specific learning outcomes, suggested learning experiences, Key Inquiry Question(s), Core competences to be developed, Pertinent and Contemporary issues developed, Values, Link to other learning areas, Community Service Learning, Non-formal activities to support learning	6
4	Multiple intelligences and differentiated learning	3
5	• Competency-based pedagogy/21st-century teaching and learning Methodologies • Suggested learning and teaching resources	3
6	**CAT 1**	2
7	• Introduction to inclusive Education • Mainstreaming Values (Value Based Education-VbE) and PCIS	3
8	Introduction to Community service learning	3
9	Competence Based Assessment	3
10	ICT integration in Education	3
11	**CAT 2**	2
12	Introduction to Comprehensive School health and safety	3
13	Preparation of professional documents	3
14	Microteaching	3
15 and 16	Assessment	2
		42

TOPIC ONE: BASIC EDUCATIONAL FRAMEWORK FOR CBC

OVERVIEW OF COMPETENCY-BASED CURRICULUM AND BASIC EDUCATION CURRICULUM FRAMEWORK

Competency-Based Curriculum (CBC)

A Competency-Based Curriculum (CBC) is an educational approach focused on developing specific skills, knowledge, and attitudes that learners need to succeed in life and work. Unlike traditional curricula that emphasize rote learning and content memorization, CBC prioritizes the ability to apply knowledge in real-world situations.

Key features of a Competence-Based Curriculum include:

1. **Learner-Centered Approach:** Focuses on the learner's needs, interests, and abilities, encouraging active participation and self-directed learning.
2. **Skills and Competencies:** Emphasizes practical skills and competencies, such as problem-solving, critical thinking, communication, and collaboration.
3. **Outcome-Oriented:** Learning is structured around achieving specific outcomes or competencies, rather than simply covering content.
4. **Continuous Assessment:** Evaluation is ongoing and based on a learner's ability to demonstrate the competencies, rather than just passing exams.
5. **Flexibility:** Allows learners to progress at their own pace, focusing on mastering competencies rather than adhering to a fixed schedule.
6. **Integration of Knowledge:** Encourages the application of knowledge across different subjects and real-life contexts.

CBC aims to equip learners with the skills needed for lifelong learning and adaptability in a rapidly changing world.

CBC Vision

An engaged, empowered and ethical citizen. The vision of a Competence-Based Curriculum (CBC) is generally centered around producing holistic, well-rounded learners who are equipped with the skills, knowledge, values, and attitudes necessary to thrive in the 21st century. While specific visions may vary depending on the country or institution implementing the CBC, the overarching goals typically include:

1. **Empowered Learners:** To develop learners who are confident, creative, innovative, and capable of critical thinking and problem-solving.
2. **Lifelong Learning:** To instill a culture of continuous learning and adaptability, enabling learners to navigate and succeed in a rapidly changing world.
3. **Holistic Development:** To foster the intellectual, emotional, social, and physical development of learners, ensuring they grow into well-rounded individuals.
4. **Real-World Relevance:** To provide education that is directly applicable to real-life situations, making learning meaningful and relevant to everyday experiences.
5. **Inclusivity and Equity:** To create an inclusive learning environment that caters to diverse learner needs, promoting equity and fairness in education.

6. **Global Competitiveness:** To prepare learners to compete and collaborate on a global stage, equipped with the skills necessary for the modern workforce.
7. **Ethical and Responsible Citizens:** To nurture responsible citizens who value integrity, sustainability, and respect for others, contributing positively to society.

The vision of CBC ultimately seeks to transform education from a focus on content delivery to a system that empowers learners to apply their knowledge and skills in meaningful ways throughout their lives.

CBC Mission

Nurturing every learner's potential. The mission of a Competence-Based Curriculum (CBC) is typically aimed at implementing an educational framework that nurtures and develops learners' abilities to become competent, responsible, and productive members of society. The specific mission may vary depending on the country or educational institution, but generally, it includes the following elements:

1. **To Develop Essential Skills and Competencies:** Focus on equipping learners with the practical skills, knowledge, and attitudes needed to succeed in both personal and professional life.
2. **To Foster Holistic Growth:** Promote the intellectual, emotional, social, physical, and moral development of learners, ensuring a balanced and comprehensive education.
3. **To Promote Learner-Centered Education:** Implement a learner-centered approach that recognizes individual differences, encourages active participation, and supports personalized learning experiences.
4. **To Encourage Lifelong Learning:** Instill a mindset of continuous improvement and adaptability, preparing learners to navigate the challenges of a dynamic and ever-changing world.
5. **To Enhance Real-World Application:** Ensure that education is relevant to real-life situations, enabling learners to apply their knowledge and skills in practical, meaningful contexts.
6. **To Support Inclusive Education:** Provide equitable learning opportunities for all students, regardless of their backgrounds, abilities, or circumstances, fostering an inclusive and supportive learning environment.
7. **To Build Ethical and Responsible Citizens:** Cultivate values such as integrity, respect, sustainability, and civic responsibility, empowering learners to contribute positively to their communities and society at large.
8. **To Promote Global Competitiveness:** Prepare learners to compete and collaborate on a global scale, equipped with the skills and competencies required in a globalized economy.

The mission of CBC is to transform education into a system that not only imparts knowledge but also builds the skills, attitudes, and values necessary for learners to thrive in all aspects of life.

Rationale for Curriculum Reforms

- **Constitution of Kenya, 2010**

- Article 53 (1) (b) states, that every child has a right to free and compulsory basic education
- Article 10, National Values and Principles of Governance
- Chapters 6, Leadership and Integrity
- Promoting Kiswahili as a national and official language and English as official languages.
- Encouraging teaching and learning of KSL and Braille.
- Developing and promoting the use of indigenous languages.
- Emphasis on communication formats and technologies accessible to persons with disabilities**.**

- **The Kenya Vision 2030**
  - Integrating early childhood into primary education
  - Reforming secondary curricula
  - Modernizing teacher training
  - Developing programmes for learners with special needs among others
- **Task Force Report chaired by Professor Odhiambo, 2012 recommended:**
  - Major curriculum reform to align it with the Constitution, 2010 and to address the dictates of Vision 2030
  - Inclusion of community service programmes in the school curriculum
- **Draft Sessional Paper on 'Reforming Education and Training in Kenya' recommends:**
  - Reforming the education sector to provide for the holistic development of learners
  - Recommends a competency based curriculum
  - Early Identification and nurturing of talents
  - Introduction of values
  - Introduction of three learning pathways at Senior Secondary School level.
- **Summative evaluation of the 8-4-4 system of education, 2009 recorded:**
  - 8-4-4 was academic and examination oriented.
  - Teachers hardly use formative assessment (assessment for learning). Assessment was limited to summative assessment (assessment of learning).
  - Graduates at secondary school level also did not acquire adequate entrepreneurial skills for self-reliance.
  - Did not provide flexible education pathways for identifying and nurturing aptitudes, talents and interests of learners early enough in order to prepare them for the world of work, career progression and sustainable development.
  - Identified skills gaps: agricultural, entrepreneurial, vocational and technical skills, innovation and creativity and ICT.
  - Recommended a needs assessment survey
- **KICD Needs Assessment Survey 2016, recommended:**
  - A curriculum reform

- A wide curriculum to cater for various talent areas to enable learners achieve their potential and be better equipped to develop themselves and the nation.
- A functional holistic curriculum that would produce an individual with spiritual, moral and social aspects with ability to live amicably with others.
- Development and acquisition competencies at every level of learning
- A flexible and comprehensive educational structure and system

- **East African Community Protocol recommendations on:**
  - National Goals of Education
  - Structure for Basic Education
  - Key Competences Learning areas/subjects
  - Learning outcomes for ECD, Primary and Secondary
  - Guidelines on assessment
- **Global Standards**
  - UNESCO IBE – recommends curriculum review every 5 years
  - Sustainable Development Goal No 4 (ensuring quality education)
  - 21st Learning (Skills and Approaches)

The 21st century learner is…

- A communicator
- A problem solver
- A critical thinker
- A collaborator
- Information and media literature
- Financially economically literate
- An innovator
- A self-directed learner

Paradigm shift – table

From (less)	**To (more)**
Content focus	Focus on the development of competencies
Rigid and prescriptive curriculum with limited flexibility	Learners' interests and special abilities should be considered in the learning process. Flexible with opportunities for specialization
Teaching where the role of the teacher is directing and authoritative	Learning where learners take charge of their learning and teacher's role is facilitating, guiding and providing resources
Teacher-centered	Learner centered
Emphasis on summative assessment and competition among learners	Emphasis on formative assessment and collaborative learning where learners share knowledge and experiences
Strict adherence to a fixed curriculum	Learners' interest are valued, which leads to identification and nurturing of talents

Learning is based on repetition	Learning is interactive and builds on what the learner already knows

Paradigm shifts in the Teacher Education Curriculum Reform

From	**To**
Focus on content knowledge	Focus on pedagogy and professional
Focus on teacher centered teaching	Focus on Learner centered pedagogy
Academic and competitive teaching	Reflective teaching and mentorship
No set standards for teacher training	Set standards articulated in the learning
Little focus on skills	Focuses on 21st Century skills
Emphasis on summative assessment	Balance of formative and summative

Pillars of CBC

- Values
- Guiding principles
- Theoretical Approaches

Values

- Love
- Responsibility
- Respect
- Unity
- Peace
- Patriotism
- Social Justice
- Integrity

How Values are integrated in the Curriculum:

- Mainly through infusion and selection of learning environment and suggested learning experiences
- Guidance and Counselling programmes
- Whole-School Approach which involves learners, teachers, support staff, Board of Management, family members, wider school community and relevant stakeholders

Guiding principles

- Opportunity
- Excellence
- Diversity and Inclusion
- Parental Empowerment and Engagement
- Community Service Learning
- Differentiated Curriculum and Learning

How the Guiding Principles have been applied in the curriculum:

- Pathways open opportunity for different categories of learners

- Stress on excellence in all teaching and learning
- Parents are engaged through extended learner activities
- Varied activities are suggested to cater for the different learning styles

Theoretical Approaches

Instructional Design Theories

- **Vygotsky's Socio-Cultural Theory**

Lev Vygotsky's Sociocultural Theory is a framework in psychology that emphasizes the fundamental role of social interaction in the development of cognition. Vygotsky, a Russian psychologist, argued that community and culture play a central role in "making meaning." His theory highlights how cultural tools, language, and interaction with more knowledgeable others (such as parents, teachers, and peers) significantly shape our learning and cognitive development.

- **Gardner's Multiple Intelligences Theory**

Howard Gardner's Theory of Multiple Intelligences, proposed in 1983, challenges the traditional view of intelligence as a single, general ability. Gardner argued that humans possess multiple intelligences, each representing different ways of processing information. His theory suggests that individuals have unique combinations of these intelligences, and that traditional IQ tests only measure a narrow range of human potential.

- **Piaget's Theory of Cognitive Development**

Jean Piaget's Theory of Cognitive Development is one of the most influential theories in the field of psychology, particularly in understanding how children develop intellectual capabilities. Piaget proposed that children go through a series of developmental stages that are marked by distinct cognitive abilities. His theory suggests that children actively construct their understanding of the world through experiences and interactions with their environment.

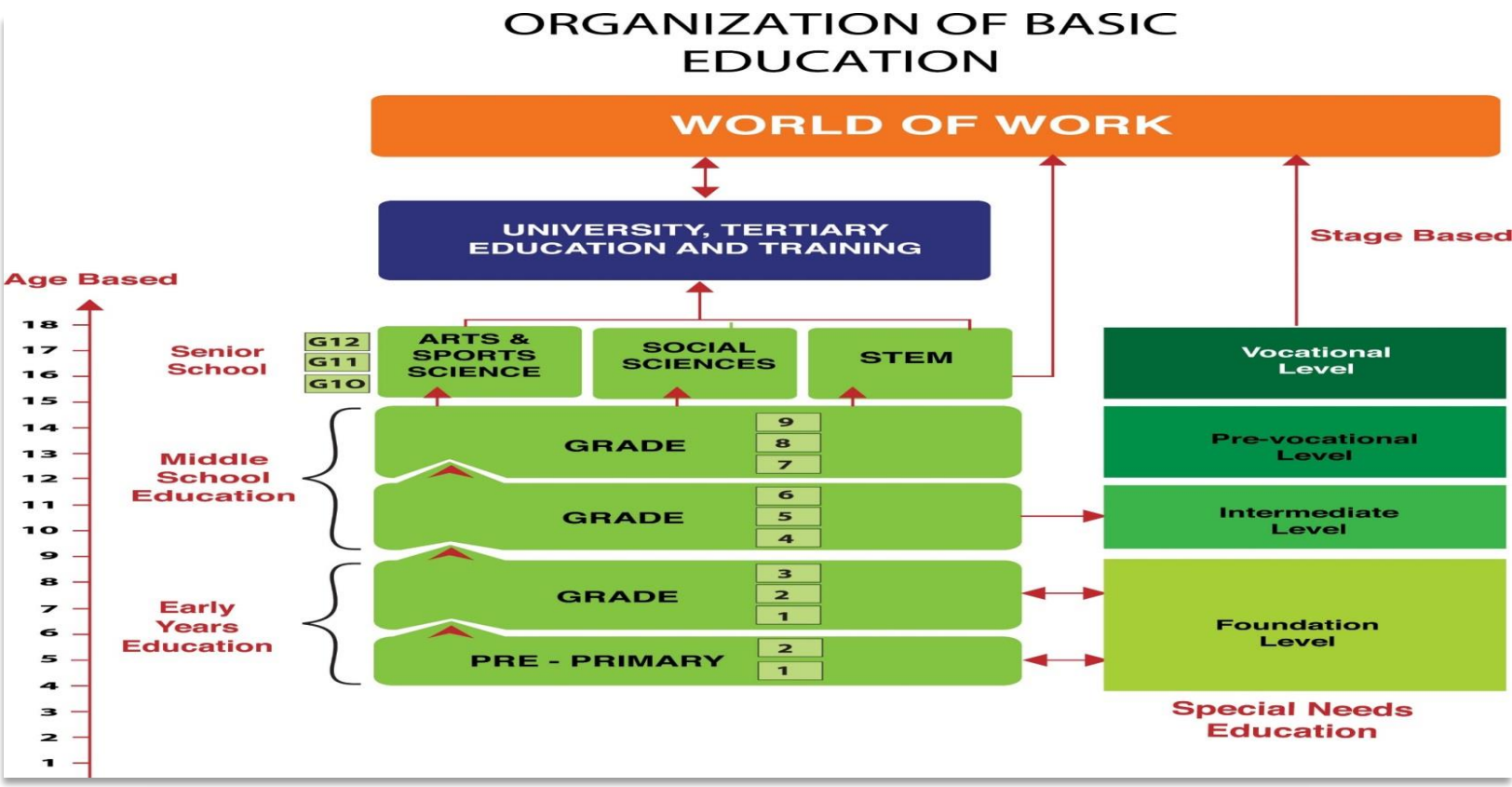
The Theories Have Been Applied in the Curriculum in:

- Designing of the curriculum and lessons.
- Selection of learning experiences
- Differentiated curriculum and learning approaches
- Selection and development of learning materials

Core Competencies

- Communication and collaboration
- Critical thinking and problem Solving
- Creativity and imagination
- Citizenship
- Digital literacy
- Learning to learn
- Self-efficacy

Course work One :With relevant examples explain the core competencies in CBE (10 marks)


ORGANIZATION OF BASIC EDUCATION
WORLD OF WORK
UNIVERSITY, TERTIARY EDUCATION AND TRAINING
Stage Based
Age Based
Senior School
G12
G11
G10
ARTS & SPORTS SCIENCE
SOCIAL SCIENCES
STEM
Vocational Level
Middle School Education
GRADE
9
8
7
GRADE
6
5
4
Pre-vocational Level
Intermediate Level
Early Years Education
GRADE
3
2
1
PRE - PRIMARY
2
1
Foundation Level
Special Needs Education
18
17
16
15
14
13
12
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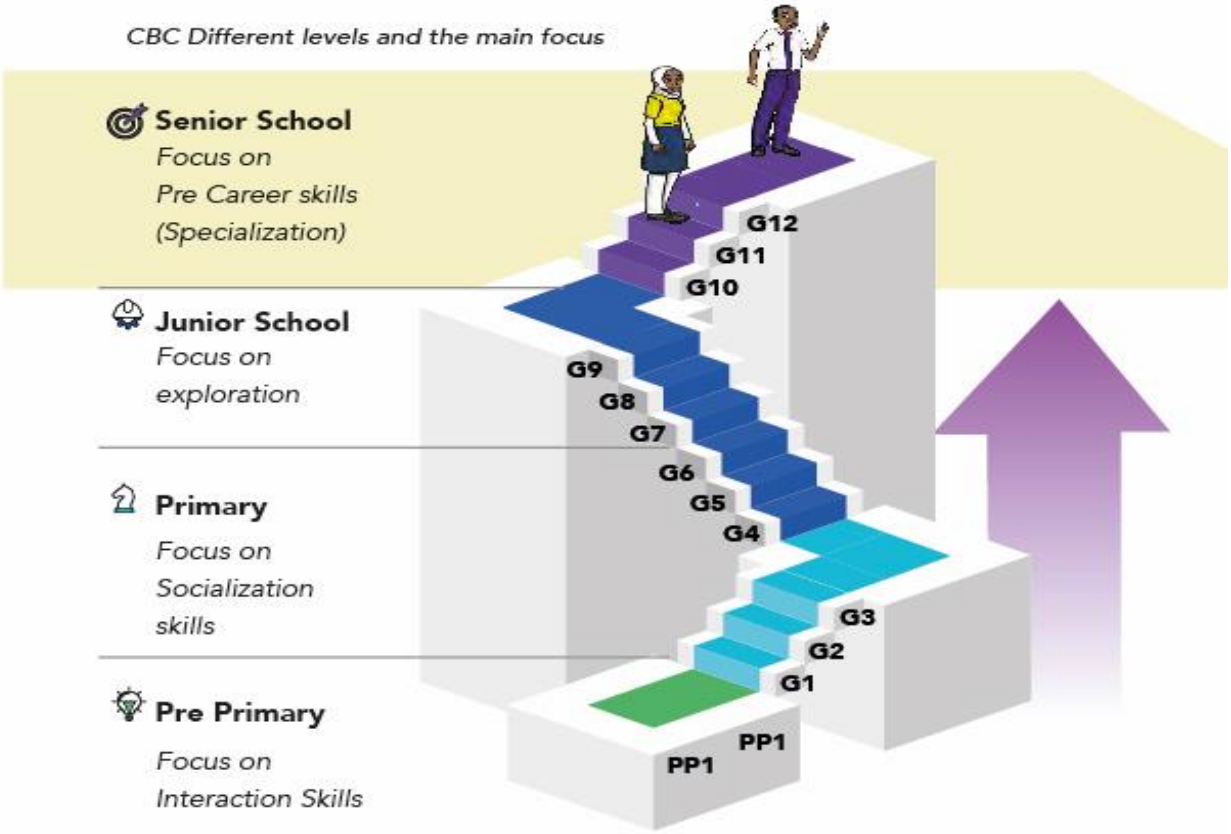

CBC Different levels and the main focus
Senior School
Focus on
Pre Career skills
(Specialization)
Junior School
Focus on
exploration
Primary
Focus on
Socialization
skills
Pre Primary
Focus on
Interaction Skills
G12
G11
G10
G9
G8
G7
G6
G5
G4
G3
G2
G1
PP1
PP1

THE PRESIDENTIAL WORKING PARTY ON EDUCATION AND REFORMS

The Presidential Working Party on Education and Reforms (PWPER) was appointed to look into issues across the Education sector from M&E reports. The issues included:

- Inadequate teacher capacity, infrastructure and cost of implementation.
- The number of learning areas was high within the available time and resources.
- Issues of content overload, overlaps, suitability of instructional materials and attainment of learning outcomes *(cost and availability).*
- Key tenets not fully understood, operationalized and realized *(Core Competencies, Values, PCI's, and PEE).*
- Implementation of CSL: overload, overlaps, cost and unclear community linkages**.**

PWPER RECOMMENDATIONS

1. A review of the curriculum to be undertaken to ensure that the content is manageable and appropriate to the level of the learners.
2. KICD to rationalize the number of learning areas and curriculum designs in terms of scope, integration of subjects and address issues of gaps, content overloads and overlaps.
3. Review activities that engage parents to ensure parents are not overburdened.
4. KICD to review CSL to reduce overload, overlaps, cost and promote community linkages.
5. Strengthen integration of the key tenets (VbE, CSL and PEE) to support successful implementation of CBC using the Whole School Approach.
6. Junior School (Grade 7-9): The learning areas not to exceed nine (9)

INTEGRATED CURRICULUM

An integrated curriculum provides learning that is synthesised across traditional subject areas and learning experiences that are designed to be mutually reinforcing each other. This creates a more cohesive and comprehensive educational experience for learners.

This approach develops the learner's ability to transfer their learning to other settings and their day-to-day activities.

During rationalization of curriculum, subjects were integrated, overlaps and overloads removed, learning experiences enriched to support implementation of key tenets of CBC. Curriculum Designs from PP 1-PP2 to Grade 9 were reviewed.

ADAPTATION OF CURRICULUM DESIGNS

- All the rationalized designs were adapted alongside the regular designs
  - Visually Impaired
  - Hearing Impaired
  - Physically Impaired
- For the Stage Based Curriculum the foundation level and intermediate levels curriculum is rationalized and reviewed.
- For the Prevocational level, the curriculum for Grade 7, 8 & 9 is rationalized and reviewed to be at par with the curriculum for regular learners**.**

Learning areas in CBC

Preprimary = 5 learning Areas

S/No	Learning areas	Number of lessons per week
1	Language Activities	**5**
2	Mathematical Activities	**5**
3	Creative Activities	**6**
4	Environmental Activities	**5**
5	Religious Activities	**3**
	Pastoral Program/Religious Instruction (PPI)	**1**
	Total	**25**

Lower primary = 7 seven learning Areas

S/No	Learning areas	Number of lessons per week
1	Indigenous Language Activities	**2**
2	Kiswahili Language Activities	**4**
3	English Language Activities	**5**
4	Mathematical Activities	**5**
5	Religious Activities	**3**
6	Environmental Activities	**4**
7	Creative Activities	**7**
	Pastoral Program/Religious Instruction (PPI)	**1**
	Total	**31**

Upper primary = 8 Learning Areas

S/No	Learning areas	Number of lessons per week
1	English	**5**
2	Kiswahili/Kenya Sign Language	**4**
3	Mathematics	**5**
4	Religious Activities	**3**
5	Science and Technology	**4**
6	Agriculture and Nutrition	**4**
7	Social Studies	**3**
8	Creative Arts	**6**
	Pastoral Program/Religious Instruction (PPI)	**1**
	Total	**35**

Junior school = 9 Learning Areas

S/No	Learning areas	Number of lessons per week
1	English	**5**
2	Kiswahili/Kenya Sign Language	**4**
3	Mathematics	**5**
4	Religious Activities	**4**
5	Social Studies	**4**
6	Integrated Science	**5**
7	Pre Technical Studies	**4**
8	Agriculture and Nutrition	**4**
9	Creative Arts and Sports	**5**
	Pastoral Program/Religious Instruction (PPI)	**1**
	Total	**41**

Senior school

The Competency-Based Curriculum (CBC) for senior school (Grades 10-12) is designed to provide students with a broad range of learning areas to prepare them for various career paths, higher education, and life skills. The CBC at the senior school level emphasizes the development of competencies, values, and practical skills that are aligned with students' interests and talents.

Key Learning Areas in Senior School (Grades 10-12) under the CBC:

The senior school curriculum is divided into three pathways, each offering specialized subjects:

1. **Arts and Sports Science Pathway**
    - **Subjects include:**
        - Performing Arts (Music, Dance, Drama)
        - Visual Arts (Fine Art, Applied Art, Craft)
        - Sports Science
        - Media Studies
        - Graphic and Interior Design
        - Fashion and Design
2. **Social Sciences Pathway**
    - **Subjects include:**
        - History and Citizenship
        - Geography
        - Religious Education (Christian Religious Education, Islamic Religious Education, Hindu Religious Education)
        - Business Studies
        - Economics
        - Sociology
        - Political Science

        - Psychology
3. **Science, Technology, Engineering, and Mathematics (STEM) Pathway**
    - **Subjects include:**
        - Mathematics (Pure and Applied)
        - Physics
        - Biology
        - Chemistry
        - Computer Science
        - Agriculture
        - Engineering Science
        - Aviation Technology
        - Geosciences
        - Health Science

Cross-Cutting Areas:

In addition to the specialized subjects within each pathway, students are also exposed to:

- **Life Skills and Values Education**
- **Community Service Learning**
- **Entrepreneurship Education**

Elective Subjects:

Students have the flexibility to choose elective subjects based on their career interests and the pathway they select.

Core Subjects:

Regardless of the pathway, certain subjects remain compulsory across all streams. These usually include:

- English
- Kiswahili
- Mathematics

How Learning is Taking Place at JS

- Learners are exposed to experiential and learner centered methods in order to acquire requisite knowledge, skills, attitudes and values. This include inquiry based learning, collaborative learning, cooperative learning, differentiated learning, among others.
- Learners are also given opportunities to engage in community service learning (CSL) activities as a way of making learning more meaningful.
- CSL is an experiential learning strategy that creates opportunities for learners to apply the knowledge, skills and values acquired in the formal dimension of their education in their community while at the same time providing age-appropriate and relevant service.
- The use of Information and Communication Technology (ICT) in curriculum delivery is emphasized to facilitate learning and enable learners develop digital literacy.

Implications on the Teacher's Competencies

The teacher:

- must be able to model the same core competencies and values as those of the learner so that s/he will be a role model for the learners,
- should be able to understand the developmental stages of the learner in Junior School,
- should be able to identify the learners' talents and create opportunities to nurture them,
- should be able to understand the age appropriate learner centered pedagogies,
- Requires adequate time for enhancing his/her competency to interpret and implement JS curricula.

Communities of Practice

- Teachers should have the ability to interact with colleagues with a view of learning from each other and improving instruction.
- Quality Teachers Circles have become the norm, teachers engage in joint activities and discussions, share information, resources and aim at generating and sustaining professional development.

PERSONAL COMPETENCIES FOR CBC TEACHERS

- **Effective Communication** (verbal, nonverbal, speaking, writing, visual and the organization of ideas into understandable structures) and listening.
- **Cooperation and collaboration-**Ability to be part of a community of practice through learning and reflection.
- **Pleasant/approachable/trustworthy personality** - Self-awareness and self-control to be able to adopt a frame of mind that would facilitate learning in the classroom
- **Patience and resilience-**Ability to withstand the pressure emanating from different aspects and challenges in and out of class
- **Intuitive perception** -This is the ability to naturally read beyond what the learner presents in their behavior, personality and interaction. Forming a rapport with the learners in and out of the classroom to facilitate learning.
- **Flexibility and ability to change and adapt.** The teacher should be able to fit in and learn from varying situations and experiences. Making informed decisions from reflective practices.
- **Ability to resolve conflict (Conflict resolution-**As a JS teacher, it will be necessary to look for ways of resolving conflicts among the learners and encourage them to resolve conflicts amicably.
- **Self-knowledge and positive image** -Self-concept skills are essential since the teacher is expected to be a role model. It entails personal grooming, mode of dress and self-confidence.

Topic 2
Interpretations of curriculum Designs
Curriculum Design

- A Curriculum design is a document that outlines the components or elements of a curriculum.
- It's a core document that guides learning institutions on how to organise learning.
- Curriculum design provides details that show the relationship between the components.
- Designs ensure the curriculum can be implemented within the time allocated and with the infrastructure in the schools.

National Goals of Education
Kenya has the following eight (8) National goals of education are as follows:
1. Foster nationalism, patriotism, and promote national unity
Kenya's people belong to different communities, races, and religions and should be able to live and interact as one people. Education should enable the learner to acquire a sense of nationhood and patriotism. It should also promote peace and mutual respect for harmonious co-existence.

2. Promote social, economic, technological, and industrial needs for national development
Education should prepare the learner to play an effective and productive role in the nation.
a) Social Needs
Education should instill social and adaptive skills in the learner for effective participation in community and national development.
b) Economic Needs
Education should prepare a learner with requisite Competencies that support a modern and independent growing economy. This should translate into high standards of living for every individual.
c) Technological and Industrial Needs
Education should provide the learner with necessary Competencies for technological and industrial development in tandem with changing global trends.

3. Promote individual development and self-fulfillment
Education should provide opportunities for the learner to develop to the fullest potential. This includes the development of one's interests, talents, and character for a positive contribution to society.

4. **Promote sound moral and religious values**
Education should promote the acquisition of national values as enshrined in the Constitution. It should be geared towards developing a self- disciplined and ethical citizen with sound moral and

religious values.

5. Promote social equity and responsibility

Education should promote social equity and responsibility. It should provide inclusive and equitable access to quality and differentiated education; including learners with special educational needs and disabilities. Education should also provide the learner with opportunities for shared responsibility and accountability through service-learning.

6. Promote respect for and development of Kenya's rich and varied cultures

Education should instill in the learner appreciation of Kenya's rich and diverse cultural heritage. The learner should value own and respect other people's culture, as well as embrace positive cultural practices in a dynamic society.

7. Promote international consciousness and foster positive attitudes towards other nations

Kenya is part of the interdependent network of diverse peoples and nations. Education should therefore enable the learner to respect, appreciate and participate in the opportunities within the international community. Education should also facilitate the learner to operate within the international community with full knowledge of the obligations, responsibilities, rights, and benefits that this membership entails.

8. Good health and environmental protection

Education should inculcate in the learner the value of physical and psychological well-being for self and others. It should promote environmental preservation and conservation, including animal welfare for sustainable development.

Level of learning outcomes

It refers to the hierarchical stages or tiers at which students are expected to demonstrate mastery of specific competencies. These levels represent the progression of learning, from basic to more complex and advanced skills, knowledge, and attitudes. The levels are designed to ensure that students build upon their previous learning as they move toward achieving higher-order competencies.

Key Aspects of Levels of Learning Outcomes in CBC

1. **Progressive Complexity**:
    - Learning outcomes are organized into levels that reflect increasing complexity and depth of understanding. As students advance through these levels, they are expected to demonstrate greater sophistication in their thinking, problem-solving, and application of knowledge.
2. **Developmental Appropriateness**:
    - Each level of learning outcomes is tailored to the developmental stage of the students. The curriculum considers the cognitive, emotional, and social

development of learners, ensuring that outcomes are achievable and appropriate for their age and educational stage.

3. **Cumulative Learning**:
    - The levels are cumulative, meaning that mastery at one level is a prerequisite for advancing to the next. Students are expected to consolidate their learning at each level before progressing, ensuring a solid foundation of knowledge and skills.
4. **Learning Outcomes Hierarchy**:
    - Typically, learning outcomes are divided into three to four levels, such as:
        - **Basic/Foundational Level**: Focuses on acquiring fundamental knowledge and skills. For example, understanding basic concepts, recalling facts, and following simple procedures.
        - **Intermediate/Proficient Level**: Involves applying knowledge and skills in familiar situations. This might include problem-solving in predictable contexts, interpreting information, and beginning to analyze.
        - **Advanced/Extended Level**: Requires the application of skills in more complex and unfamiliar situations. It involves critical thinking, creativity, synthesis of ideas, and making judgments based on evidence.
        - **Expert/Mastery Level** (if applicable): At this highest level, students demonstrate deep understanding, mastery, and the ability to transfer skills and knowledge to new and varied contexts. They are capable of independent, self-directed learning.
5. **Assessment at Each Level**:
    - Assessments are aligned with the levels of learning outcomes, ensuring that students are evaluated based on the competencies expected at their current level. Formative assessments guide learning at lower levels, while summative assessments measure proficiency and readiness to advance.
6. **Learning Continuum**:
    - The levels of learning outcomes create a continuum of learning, where students progress at their own pace. This allows for differentiation in instruction, accommodating students who may need more time at certain levels or those ready to move ahead more quickly.

Example of Levels of Learning Outcomes

For a competency in problem-solving in mathematics, the levels might be:

1. **Basic Level**: Identify and understand basic mathematical problems.
    - Example Outcome: "Students can recognize and define simple mathematical problems in addition and subtraction."
2. **Intermediate Level**: Apply mathematical concepts to solve routine problems.
    - Example Outcome: "Students can solve addition and subtraction problems involving multiple steps."
3. **Advanced Level**: Analyze complex mathematical problems and develop solutions.
    - Example Outcome: "Students can analyze word problems and apply appropriate mathematical operations to find solutions."

4. **Expert Level** (if included): Transfer problem-solving strategies to new and unfamiliar contexts.
    - Example Outcome: "Students can develop and test hypotheses to solve novel mathematical problems and explain their reasoning."

Importance of Levels of Learning Outcomes in CBC

1. **Structured Learning Path**:
    - Levels of learning outcomes provide a structured pathway for students, ensuring that learning is systematic and that competencies are developed progressively.
2. **Clear Expectations**:
    - Clearly defined levels help students and teachers understand what is expected at each stage of learning, making it easier to set goals and measure progress.
3. **Customized Instruction**:
    - Teachers can use the levels to differentiate instruction, providing appropriate challenges and support based on each student's current level.
4. **Focus on Mastery**:
    - By emphasizing mastery at each level before moving on, the CBC ensures that students have a deep and thorough understanding of the material, rather than simply progressing through the curriculum without fully grasping the concepts.

In summary, Levels of Learning Outcomes in a Competence-Based Curriculum provide a framework for the progressive development of competencies, ensuring that students build knowledge and skills in a systematic, developmentally appropriate, and cumulative manner.

Learning Area general outcomes

It refers to the broad, overarching goals that define what students should achieve in a particular subject or learning area by the end of a specific educational stage (such as a grade level, key stage, or entire course). These outcomes summarize the essential knowledge, skills, attitudes, and values that the curriculum aims to develop in students within that learning area.

Key Features of Learning Area General Outcomes

1. **Broad Objectives**:
    - General outcomes are not tied to specific lessons or units but rather reflect the overall objectives of the entire learning area. They encompass the key competencies that students are expected to develop across the subject.
2. **Integration of Competencies**:
    - These outcomes integrate various competencies, including cognitive (knowledge and understanding), psychomotor (skills and abilities), and affective (attitudes and values) domains. This ensures a holistic development of the learner.
3. **Guiding Instruction**:
    - General outcomes guide the planning and delivery of instruction. They help educators design lessons, select content, and choose teaching strategies that align with the broader goals of the curriculum.
4. **Assessment Benchmarks**:

    - These outcomes serve as benchmarks for assessing student progress. They provide a framework for evaluating whether students have achieved the intended competencies by the end of the learning period.
5. **Contextual Relevance**:
    - Learning Area General Outcomes are often contextualized to reflect the specific needs, culture, and priorities of the educational system or community. They are designed to be relevant to students' lives and applicable to real-world situations.

Example of Learning Area General Outcomes

For a learning area like **Science** at the end of a key stage (e.g., elementary school), general outcomes might include:

1. **Understanding Scientific Concepts**:
    - **Outcome**: "Students demonstrate an understanding of key scientific concepts, including the structure and function of living organisms, physical properties of materials, and basic principles of energy and force."
2. **Applying Scientific Methods**:
    - **Outcome**: "Students use scientific methods to explore questions, conduct experiments, and draw conclusions based on evidence."
3. **Critical Thinking and Problem Solving**:
    - **Outcome**: "Students apply critical thinking and problem-solving skills to analyze scientific information and solve real-world problems."
4. **Environmental Awareness and Responsibility**:
    - **Outcome**: "Students recognize the importance of environmental stewardship and demonstrate responsible behaviors in relation to the environment."
5. **Communication of Scientific Ideas**:
    - **Outcome**: "Students effectively communicate scientific ideas and findings through various forms, including written reports, presentations, and discussions."

Importance of Learning Area General Outcomes in CBC

1. **Curriculum Coherence**:
    - General outcomes ensure coherence across the curriculum by providing a consistent focus for teaching and learning in a particular subject area. This helps in aligning lesson plans, instructional materials, and assessments.
2. **Clarity of Purpose**:
    - They provide clarity of purpose for both educators and students. Teachers understand the ultimate goals they are working towards, and students know what is expected of them by the end of the course or grade level.
3. **Focus on Competency Development**:
    - By emphasizing broad competencies rather than just content knowledge, general outcomes support the development of skills that are transferable across different contexts and subjects, fostering lifelong learning.
4. **Assessment and Accountability**:
    - General outcomes serve as key points of reference for assessing student achievement and ensuring that the curriculum is effectively meeting its goals.

They are also useful for reporting progress to stakeholders, such as parents and administrators.

5. **Flexibility in Instruction**:
    - While they provide a clear framework, general outcomes allow for flexibility in how teachers achieve these goals. Educators can adapt their instructional strategies to meet the needs of their students while still aiming for the same broad objectives.

In summary, Learning Area General Outcomes in a Competence-Based Curriculum define the broad, essential goals that students should achieve within a specific subject area. They guide instruction, support assessment, and ensure that education is focused on developing well-rounded, competent individuals.

Essence statement

An **Essence Statement** in a Competence-Based Curriculum (CBC) is a concise summary that captures the core purpose and key elements of a particular subject, course, or unit. It provides a clear overview of what is essential for students to learn, understand, and be able to do by the end of the instructional period. The essence statement is typically aligned with the competencies that the curriculum aims to develop in students.

Key Features of an Essence Statement in CBC

1. **Core Purpose**:
    - The essence statement articulates the fundamental purpose of the subject or unit. It explains why the content is important and how it contributes to the overall educational goals of the curriculum.
2. **Key Competencies**:
    - The statement highlights the specific competencies that students are expected to develop. These competencies could include knowledge, skills, attitudes, and values that are central to the subject.
3. **Learning Outcomes**:
    - It outlines the intended learning outcomes, focusing on what students should know, understand, and be able to do after completing the unit or course.
4. **Relevance and Application**:
    - The essence statement often emphasizes the relevance of the subject matter to real-life situations. It shows how the competencies acquired can be applied in practical contexts, helping students see the value of their learning.
5. **Clarity and Brevity**:
    - The essence statement is concise, usually a few sentences long, and written in clear, accessible language. It should be easily understandable by educators, students, and other stakeholders.

Example of an Essence Statement

For a unit on Environmental Science in a Competence-Based Curriculum, an essence statement might look like this:

"The study of Environmental Science empowers students to understand the interdependence between human activities and the natural world. By developing competencies in critical thinking,

problem-solving, and responsible citizenship, students learn to make informed decisions that promote environmental sustainability and contribute to the well-being of their communities."

Importance of Essence Statements in CBC

1. **Guiding Instruction**:
   - The essence statement serves as a guide for educators in planning and delivering instruction. It helps teachers focus on the most important aspects of the curriculum and ensures that all activities are aligned with the intended competencies.
2. **Clarifying Expectations**:
   - For students, the essence statement clarifies what is expected of them. It helps them understand the purpose of their learning and the competencies they need to develop.
3. **Ensuring Alignment**:
   - Essence statements help ensure that all components of the curriculum—such as learning activities, assessments, and instructional materials—are aligned with the desired outcomes.
4. **Facilitating Communication**:
   - These statements are useful tools for communicating the goals and focus of the curriculum to parents, administrators, and other stakeholders, fostering a shared understanding of what the educational program aims to achieve.

In summary, the essence statement in a Competence-Based Curriculum distills the most important aspects of a subject or unit, guiding both instruction and learning towards the development of key competencies in students.

COURSE WORK @ 2; PROVIDE AN ESSENCE STATEMENT FOR A UNIT IN YOUR AREA OF SPECIALISATION (5 MARKS)

Components of the CBC **Curriculum Design**

Strand	**Sub Strand**	**Specific Learning Outcomes**	**Suggested Learning Experiences**	**Key Inquiry Question(s)**
Core competences to be developed:				
Values:				
Pertinent and Contemporary Issues(PCIs):				
Link to other subjects:				

Strands and sub strands

Strand: A strand is a broad category or domain within a curriculum design that represents a major area of learning. Strands typically encompass a group of related knowledge, skills and attitudes.

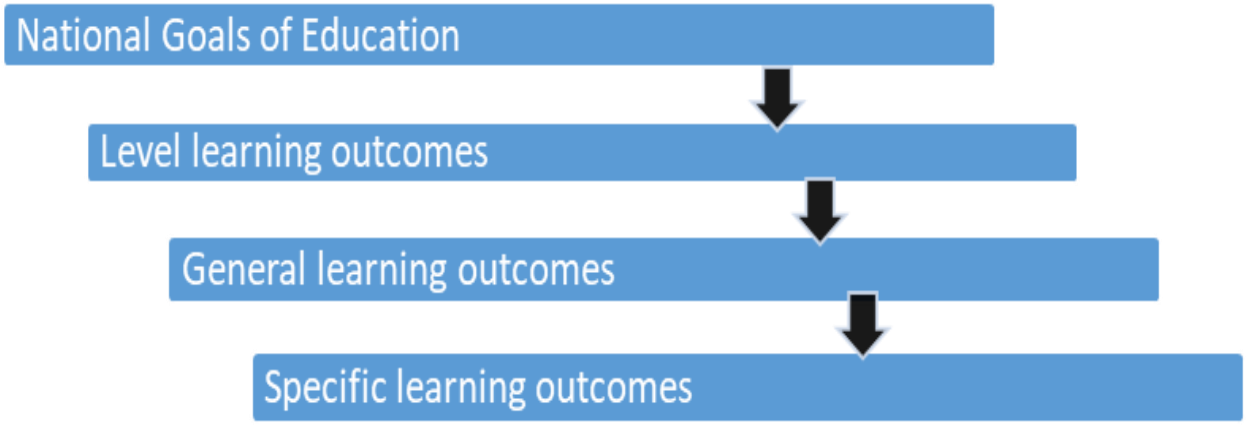
In the Pre-Technical Studies curriculum design, there are five strands: Safety, Materials, Tools, Drawing, and Energy Resources. Each strand represents a significant content area within the subjects.

Sub strand: A sub-strand is a more specific subset or subdivision within a larger strand. It further breaks down the content or competency areas into more focused and manageable components.

Specific Learning outcomes

1. The Competency Based Curriculum places emphasis on learners showing or demonstrating that they have learnt.
2. Learning outcomes are the end result or what the learner is able to demonstrate upon completion of a lesson, sub strand, strand or level. It is the evidence that learning has taken place.
3. Learning outcomes are the evidence that learning experiences have made a change that learners manifest.
4. The change or transformation in learners is manifested through new knowledge, skills, values & attitudes.

Progression of Learning Outcomes



Designing learning outcomes

a) A good learning outcome begins with an action verb, followed by the object of the verb and a phrase that gives the context.
b) Preferably only one verb is used in a learning outcome.
c) The learning outcomes should be observable.
d) Learning outcomes are amenable to assessment.
e) Avoids vague terms like *know, understand, learn, be familiar with, be exposed to, be acquainted with, and be aware of.*

Learning Outcomes must develop the following:

- **KNOWLEDGE**

    - Acquisition of facts, ideas, concepts; their application, synthesis to get new ideas and evaluation of the information
- **SKILLS**
    - Using the knowledge acquired, a learner is able to apply it to perform certain tasks against a given standard. The standard to which a learner should perform the task is referred to as a competence.
- **VALUES**
    - Desirable beliefs and personal attributes that a learner acquires through knowledge and skill that influence performance.

The 'verb' in a learning outcome

Verbs such as *identify, name, list*... illustrate knowledge.

By the end of the lesson the learner should be able to:

a) outline the importance of water conservation in farming
b) construct water retention structures to conserve surface runoff
c) appreciate the role of water conservation measures in farming

The verbs represent the three domains of learning including **knowledge, skills and attitudes.**

The 'object' in a learning outcome

The learning outcome also has an object, which answers the question **"what"**

a) outline the importance of water conservation in farming
b) construct water retention structures to conserve surface runoff
c) appreciate the role of water conservation measures in farming

The 'context' in a learning outcome

The learning outcome also has a context, which answers the question **"why", "where", "how" and "when"**

a) outline the importance of water conservation in farming
b) construct water retention structures to conserve surface runoff
c) appreciate the role of water conservation measures in farming

Formulating specific learning outcomes

Bloom's Level	**Key Verbs (keywords)**	**Example Learning outcome**
Create	design, formulate, build, invent, create, compose, generate, derive, modify, develop.	Design a garment for a new born baby
Evaluate	choose, support, relate, determine, defend, judge, grade, compare, contrast, argue, justify, support, convince, select, evaluate.	Choose foods rich in vitamin C in their locality
Analyse	classify, break down, categorize, analyse, diagram, illustrate, criticize, simplify, associate.	Differentiate between macro and micro nutrients found in foods.

Apply	calculate, predict, apply, solve, illustrate, use, demonstrate, determine, model, perform, present.	Calculate BMI for promoting health.
Understand	describe, explain, paraphrase, restate, give original examples of, summarize, contrast, interpret, discuss.	Describe materials used to build homes
Remember	list, recite, outline, define, name, match, quote, recall, identify, label, recognize.	Recite the national anthem of Kenya

Suggested learning experiences

What are learning experiences?

- Learning experiences are activities or tasks that a learner engages in to develop the desired knowledge, skills, values and attitudes.
- Learning experiences focus on developing the learner's cognitive, psychomotor and emotional domains.
- They are interactions between the learner and external conditions in the environment.

Types of Learning Experiences

The learning experiences could be:

- Formal :Organized/structured, guided and intentional
- Non-formal: Takes place outside formal learning environments but within some kind of organizational framework.
- Informal: Takes place outside the classrooms/school and arises from the learner's involvement in activities that are not undertaken with a learning purpose in mind. Involuntary daily life incidental

Factors to consider when selecting learning experiences

- Relevance to intended learning outcomes
- Time required for the experience
- Developmental age level
- Learning resources
- Safety

How can the teacher select learning experiences?

Selection of learning experiences should be based on:

Specific learning outcomes. The learner should achieve the intended specific outcomes after active engagement in the selected learning experiences.

Flexibility- the teacher has freedom to develop appropriate learning experiences besides the **suggested learning experiences** in the curriculum designs.

However, in some subjects, the suggested (L.E.,s) specify the scope of content to be covered, hence limits curriculum implementers use of the suggested (LEs) only. For example, in G-8 Integrated Science;

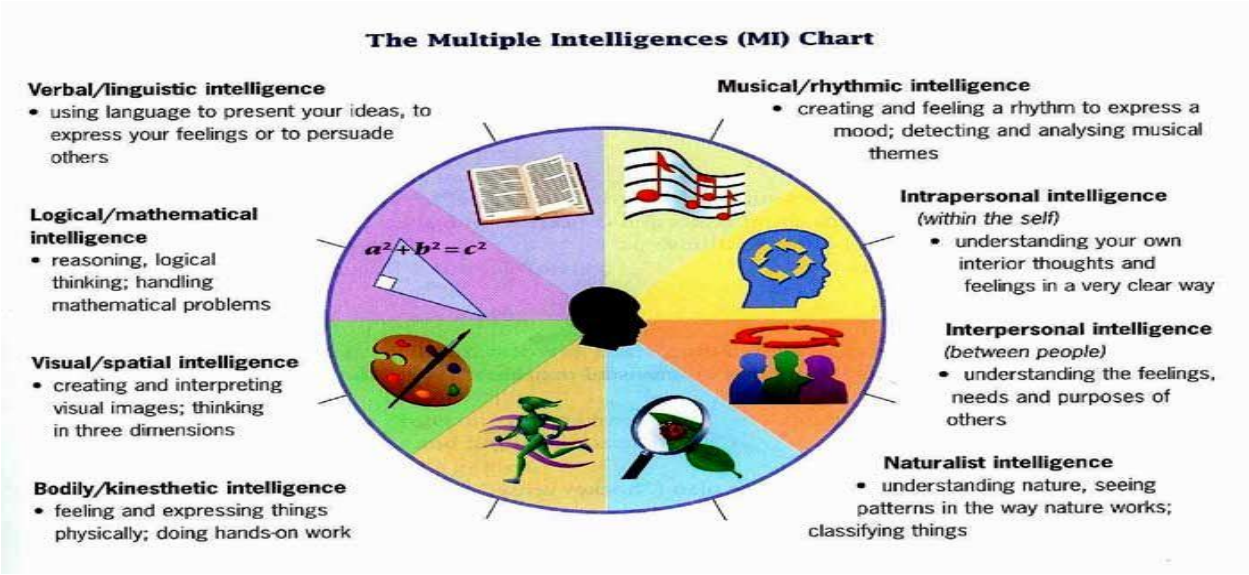
What Informs Learning Experiences?

- Learning Experiences

- **Prior Knowledge**
- **Learning Styles**
- **Differentiated Learning**
- **Multiple intelligence theory**

Learning experiences may vary depending on the different **learning styles,** needs, classroom conditions, and teaching approaches.

- **Differentiated learning** recognizes that learners learn differently and should be given an opportunity to the use of a variety of learning experiences,
- **The experiences** must be age appropriate, varied, exciting and cater for learners with special needs and disabilities.
- **Learners Prior Knowledge**, comprise learners' preconceptions, misconceptions and alternative conceptions about a sub strand /concept.



Key Inquiry Question(s)

- Inquiry-based learning is one of the critical methodologies in the competency-based curriculum. It involves asking questions.
- Such questions help the learner retain information and invariably encourage them to investigate concepts under study by asking further questions about what they are learning.

What kind of Questions are KIQs?

- These are questions that help to focus on learning.
- Questions that probe for deeper meaning and set the stage for further questioning
- Foster the development of critical thinking skills and higher order capabilities such as problem solving.
- Questions that originate from curiosity and engagement with subject matter

What Makes a Good KIQ

- It is **open-ended, non-judgmental, meaningful and purposeful** with an aim to allow learners to explore ideas.
- Is **thought-provoking and intellectually engaging**, often sparking **discussion and debate**.
- Encourages **collaboration** amongst learners, teachers, and the community.
- **Integrates technology** to support the learning process particularly through **the search for more information**.
- Call for **higher-order thinking**, such as **analysis, inference, evaluation, prediction**. It cannot be effectively answered by recall alone.
- Point toward **important, transferable ideas** within and even across subjects or learning areas.
- **Raise additional questions** and **sparks further inquiry**.
- Requires support and justification, not just an answer.

Question Words to Use

Utilize the six typical question words:

- Why?
- How?
- Who?
- Where?
- When?
- What?

Core competencies to be developed

Mainstream Core Competencies

- The competency-based approach to learning is intended to develop in the learner: *'the ability to apply appropriate knowledge and skills to successfully perform a function'*
- Learners are enabled to make connections between the knowledge they acquire and how they can use it meaningfully within a given context.
- Learning therefore becomes more purposeful and relevant to real life situations.

Core competencies

1. **Communication and Collaboration**

- Embed learning experiences that will make learning take place collegially, in a group or a team
- Present opportunities for learners to:

➢ express themselves orally and in writing
➢ listen attentively
➢ argue out an opinion and defend his or her views
➢ share knowledge/resources with others and seek the same from them
➢ contribute to the team's objectives
➢ Justify course or action taken.

2. **Critical Thinking and Problem Solving**

- Structure learning around situations that will require learners to use logic and evidence to arrive at conclusions or solve problems
- Create scenarios for learners to:
  - demonstrate objective ideas, opinions and views
  - use evidence to draw conclusions
  - innovate to solve problems in line with their age
  - explore options/new ways of doing common chores
  - generate and implement ideas individually and/or collectively to seek solutions to a contextual problem
  - demonstrate open mindedness to new ideas and opinions
  - adopt multifaceted dimensions and make multiple attempts to solving a contextual issue
  - make critical observations and deduce objective conclusions
  - make critical hypothesis and attempt resolutions to a problem
  - give objective ideas to solving a contextual problem.
  - seek assistance in critical needs.

3. Creativity and Imagination

- Exploit learning activities as avenues for learners create new ideas that result in products that add value to their lives and to the lives of those around them.
- Diversify the activities to take on board various learners' inclination to:
  - Form and communicate / present idea (writing, sketching, gestures)
  - Translate ideas to real items (drawing, sculpture, model, design)
  - Present **multiple dimensions** as a single idea
  - **Compile** other people's ideas to a concrete image
  - Compile ideas to **develop a concept**
  - **Patch-up** ideas to a concrete course/solution/concept
  - **Analyse** a broad idea into component ideas
  - **Innovate** a model/item from own or others ideas

4. Citizenship

- Create a learning environment that will require learners to demonstrate their rights, privileges, and duties as citizens.
- Engage them in activities requiring them to:
  - Value a sense of identity with others
  - Uphold identity with peers
  - Respect and uphold rights of others
  - Operate within their own rights
  - Responsibly claim their own rights and privileges
  - Undertake duties and obligations
  - Express own belonging among others
  - Seek peace while resolving conflicts with others
  - Demonstrates tolerance in resolving controversies
  - Demonstrates some level of understanding when own rights/privileges are infringed.

5. Learning to Learn

- Regularly remind learners that learning is a continuous process that requires personal organization, effective management of time and information, both individually and in groups.
- Provide tasks that will challenge learners to:
  - Show openness to new ideas
  - Critique his or her own ideas
  - Use one idea as a learning experience to a new idea
  - Continuously demonstrate personality progress
  - Use new opportunities as a learning experience to expound knowledge and skills
  - Progressively adopt new value, beliefs and opinion structures
  - Persistently pursue new targets and more challenging goals
  - Work progressively to overcome existing obstacles
  - Seek assistance and guidance to overcome persistent obstacles
  - Adopt and assimilate new knowledge and skills to overcome contextual challenges.

6. Self-efficacy

- Use the learning environment to instil in learners the belief about their capabilities to perform assigned tasks, prospects for accomplishment and personal well-being.
- Develop assignments that give learners chances to:
  - Present ideas with confidence
  - Demonstrate sense of assurance and trust
  - Present self-interest, group interest and /or defend opinions politely
  - Volunteer to undertake challenging tasks
  - Courageously volunteer to take group leadership
  - Demonstrate intrinsic self-motivation
  - Demonstrate self-awareness, responsibility, resource care and age-related chore management
  - Confidently protect and conserve personal and group resources

7. Digital Literacy

- Expose learners to a wide range of content and devices to equip them with knowledge, skills and behaviours which are effective and safe for digital literacy.
- Explore opportunities in and out of school for learners to progressively:
  - Use digital device communication networks,
  - Engage in online communication and social networks,
  - Become aware of and adhering to ethical behaviour protocols,
  - Get information about societal issues through digital media,
  - Search, evaluate and use information channelled through digital platforms

Conclusion

"Give the pupils something to do, not something to learn; and the doing is of such a nature as to demand thinking; learning naturally results" – John Dewey

Pertinent and contemporary issues developed

- During the National survey that was conducted in 2016 to inform development of Competency Based Curriculum, a number of issues were identified and found pertinent for inclusion in the curriculum.
- This together with findings from other Educational works and best practices in other countries informed identification of PCIs to be mainstreamed in the CBC curriculum
- Learners are faced with a myriad of challenges and issues owing to the social-cultural and economic dynamics in the society.
- These issues are referred to as Pertinent and Contemporary Issues (PCIs) in education because of the impact they can have on the learner and learning.
- PCIs have the potential of affecting learner's psychosocial well-being, access to education, retention, transition, completion and ultimately advancement in life after school.

CATEGORIES OF PCIs

In CBC, the following categories of PCIs have been identified for mainstreaming in the curriculum:

a) Health Promotion Issues (HPIs)
b) Life Skills and Human Sexuality
c) Socio-Economic and Environmental Issues
d) Parental Empowerment and Engagement (PE&E)
e) Citizenship Education
f) Learner Support Programs (LSPs)

HEALTH PROMOTION ISSUES

- Drug & substance use
- Mental health
- Alcohol & substance
- Harmful cultural practices
- Frist aid
- Non-communication diseases
- Communicable disease
- HIV & Aids

Life skills and human sexuality

- Assertiveness
- Negotiation
- Peer pressure resistance
- Problem solving
- Critical & creative thinking
- Managing stress
- Coping with emotions
- Self esteem
- Self-awareness

Social economic & Environmental issues

- Animal welfare
- Environment conservation: Bio-diversity, disaster, Risk reduction, pollution, Environment & technology
- Gender issues
- Safety & security: Fire safety, road safety, radicalization & violent extremism, cyber bullying prevention

Parental Empowerment and Engagement

- Participation in school events
- Learning environment
- Role modelling
- Positive discipline
- Volunteering
- Resource mobilization
- Parental involvement in decision making
- Human growth and development stages

Citizenship

- Children's rights
- Child care & protection
- Peace education
- Ethnic & racial relations
- Social cohesion
- Good governance

Mainstreaming PCIs is done through:

Formal Dimension

A relevant and suitable sub strand or sub theme in a learning area/subject is identified where PCIs can appropriately be incorporated and taught alongside the subject matter**.** In every sub strand in the design, there is a component referred to as 'link to PCIs'. This is where PCIs to be mainstreamed in the sub strand are indicated in the curriculum designs

Non-formal Dimension

All the PCIs mainstreamed in the formal dimension of the curriculum, can also be mainstreamed in Non Formal Programs such as games, sports, clubs and societies among others.

Learner Support Programs (LSPs)

- They entail the various structured learning experiences that complement formal education. They aim at linking education to the learners' everyday life, equipping them with competences necessary for molding responsible, adaptive, motivated and innovative individuals.
- LSP programs (also classified as a PCIs) include Guidance services, Career guidance, counselling services, Peer education, Mentorship, Coaching, Clubs and societies, Sports and games and Chaplaincy.

- The overall aim of the LSPs is to provide non-formal learning opportunities that facilitate learners' personal, social, physical, cognitive and career development.

Examples of learner support programmes include:

- Career guidance and services
- Peer education and mentorship
- Chaplaincy
- Clubs and societies
- Sports and games
- Counselling
- Guidance services

Values

- Values are defined as standards that guide an individual/group on how to respond or behave in a given circumstance.
- The overall goal of integrating values in the curriculum is to nurture values in learners to become ethical citizens for positive, holistic and harmonious living in a society.

Core Values in the BECF

- Love
- Responsibility
- Respect
- Unity
- Peace
- Patriotism
- Social Justice
- Integrity

Mainstreaming of Values in the Curriculum

Formally

Selection of learning experiences where the learner will be guided to nurture the targeted value e.g. during group work, debates, dramatization, talks, sharing of resources, concepts like etiquette

Non -formally

Guiding learners to exhibit values within club, Societies, Sports activities etc – conducing value based or moral activities.

Informally

The school community should create 'a culture' of expected social norms that everyone should model. Informal interactions can be through talking walls and trees, modelling proper dressing, polite language, sharing resources among others

Conclusion

Effective mainstreaming of PCIs and Values call for deliberate effort by teachers to ensure most, if not all of them are addressed during the learning experiences. Creativity is required to ensure that the PCIs & values are effectively mainstreamed. A teacher needs to be keen and sensitive to the contextual issues as they mainstream PCIs

Link to other learning areas

Community Service learning

The new Competency Based Curriculum (CBC) seeks to guide learners to become engaged, empowered and ethical citizens who will contribute to creation of a socially, economically, and politically stable society.

The vision of CBC is to nurture every learner's potential based on a strong foundation of knowledge, competencies and values.

As part of CBC, learners participate in a new learning area called Community Service Learning (CSL).

In CSL, learners work on real-life problems making academic learning relevant and at the same time developing their core competencies and values.

What is CSL?

- Is an educational approach that integrates service in the community with intended learning activities
- Educational institutions and the community work together towards outcomes that are mutually beneficial.
- Is a learning approach where learners engage in meaningful learning experiences and develop through active participation in thoughtfully organized projects that meet actual community needs
- Is an experiential learning strategy that integrates classroom learning and community service to enable learners to reflect, experience and learn from the community

Community Service-Learning vs Community Service

- **CSL** is an experiential learning strategy that integrates classroom learning and community service to enable learners reflect, experience and learn from the community. CSL activities are embedded as learning experiences to help learners develop the specified competencies.
- **Community Service** can be performed at any time for any reason.
- **Examples**
- Learners participate in community clean up every morning.
- Grade 3 learners study waste collection, sorting, disposal and participate in a clean-up project in the local market.

Benefits of CSL

- Personal efficacy, personal identity, spiritual growth and moral development
- Improved interpersonal skills: works well with others. Builds leadership and communication skills
- Helps identify and nurture learners talents, identity their strengths and address their weaknesses
- Improved inter-cultural understanding and accommodation
- Improved social responsibility and citizenship skills
- Increased involvement in community service after school
- Community benefits from the services provided by the learners

- Community learns from the school and learners
- Community benefits from the resources in the school
- Learners build connections with professionals and community members for learning and career opportunities
- Helps learners to identify their talents and areas of interest early enough.

CSL LEARNING OUTCOMES

By the end of the course the learner should be able to:

- Apply acquired competencies in the community
- Actively participate in relevant projects within the school and wider community environment for mutual benefit and learning
- Apply basic research skills to address identified community needs
- Participate in local, national and global communities as responsible citizens
- Appreciate diversity for harmonious existence within the wider community
- Communicate effectively using appropriate modes in diverse contexts within the community
- Apply sound financial and resource management skills
- Conduct oneself responsibly and relate appropriately with other members of the community while upholding desirable values
- Establish sustainable enterprises that benefit self and the community.

CSL

- CSL addresses student **learning** and identified real **needs** in the community.
- We all belong to a community.
- We exist to serve the community.
- Education should impart learners with knowledge, skills, values and positive attitude to serve the community.
- CSL seeks to help us identify and nurture learner's areas of strength, interest, gifts and talents. It provides room for learners to explore before they finally settle on their areas of strength and interest.

Example of CSL

Grade 3 Pupils Undergoing KNEC Assessment Gift Elderly Man Blanket, Firewood

Grade 3 Pupils Undergoing KNEC Assessment Gift Elderly Man Blanket, Firewood

Conclusion on CSL

- CSL combines classroom learning and community service
- CSL gives learners the opportunity to interact and work with community members to solve identified local problems
- The integration of theory and practice enables learners to experience, think critically, reflect and learn from the projects they undertake.
- In primary and junior secondary levels, CSL is integrated.
- In Senior School, CSL is a separate learning area

Non-formal activities to support learning

Non-formal activities play a significant role in supporting learning in a Competence-Based Curriculum (CBC). These activities take place outside the traditional classroom setting and are designed to complement formal education by providing students with practical, hands-on experiences that enhance their skills, knowledge, and attitudes. Below are some examples of non-formal activities that can support learning in a CBC.

1. Clubs and Extracurricular Activities

- **Science Clubs**: Students can participate in activities like science fairs, experiments, and innovation challenges, allowing them to apply scientific principles and develop problem-solving and critical thinking skills.
- **Debate and Public Speaking Clubs**: These clubs help students improve their communication skills, critical thinking, and confidence by engaging in debates, public speaking competitions, and discussions on various topics.

- **Art and Drama Clubs**: Students can explore their creativity through painting, drawing, sculpture, acting, and theater production, helping them develop artistic skills and self-expression.
- **Sports Teams and Physical Activities**: Involvement in sports and physical education fosters teamwork, leadership, discipline, and physical fitness, contributing to the holistic development of students.
- **Music and Dance Clubs**: Participation in music bands, choirs, and dance groups helps students develop their musical talents, rhythm, coordination, and cultural appreciation.

2. Community Service and Service-Learning Projects

- **Volunteering**: Students can volunteer at local organizations, such as hospitals, animal shelters, or community centers, where they can apply their skills in real-world contexts while developing empathy, social responsibility, and a sense of community.
- **Environmental Conservation Projects**: Activities like tree planting, clean-up drives, and recycling initiatives help students learn about environmental stewardship and sustainability, while actively contributing to their communities.
- **Social Entrepreneurship**: Encouraging students to create and implement projects that address social issues, such as starting a small business or creating awareness campaigns, can help them develop entrepreneurial skills and social awareness.

3. Field Trips and Educational Excursions

- **Visits to Museums and Historical Sites**: Field trips to museums, historical landmarks, and cultural centers allow students to connect their learning with historical and cultural contexts, deepening their understanding of various subjects.
- **Nature Walks and Environmental Tours**: Guided nature walks, visits to national parks, and environmental tours provide opportunities for students to learn about ecology, biodiversity, and conservation in a hands-on setting.
- **Industry Visits**: Tours of factories, farms, tech companies, or research institutions help students understand how theoretical concepts are applied in the real world, exposing them to potential career paths and industries.

4. Workshops and Skill Development Sessions

- **Coding and Technology Workshops**: Workshops on coding, robotics, and digital literacy equip students with essential 21st-century skills, fostering their technological competence and problem-solving abilities.
- **Crafts and DIY Workshops**: Hands-on sessions on crafts, woodworking, or DIY projects help students develop practical skills, creativity, and resourcefulness.
- **Career Guidance and Life Skills Workshops**: Sessions focused on resume writing, interview skills, financial literacy, and time management prepare students for future career opportunities and everyday life challenges.

5. Competitions and Challenges

- **Academic Competitions**: Participation in math Olympiads, spelling bees, essay contests, and quiz competitions allows students to test their knowledge and skills in a competitive environment, fostering excellence and motivation.

- **Innovation and Entrepreneurship Challenges**: Students can engage in hackathons, business plan competitions, and invention contests, where they develop and present innovative solutions to real-world problems.
- **Art and Creative Contests**: Art exhibitions, photography contests, and creative writing competitions encourage students to showcase their talents and express their ideas creatively.

6. Peer Tutoring and Mentorship Programs

- **Peer Tutoring**: Older or more advanced students can tutor their peers in subjects they excel in, reinforcing their own knowledge while helping others improve their understanding of key concepts.
- **Mentorship Programs**: Pairing students with mentors from various professions or community leaders provides them with guidance, career advice, and role models to look up to, helping them navigate their academic and personal growth.

7. Cultural and Heritage Activities

- **Cultural Festivals and Celebrations**: Organizing and participating in cultural festivals, traditional dance performances, and heritage day events allow students to explore and appreciate their cultural heritage, promoting cultural competence and identity.
- **Language Immersion Programs**: Activities such as language camps, exchange programs, or cultural immersion experiences help students develop linguistic skills and cultural awareness, particularly in multilingual settings.

8. Environmental and Outdoor Education

- **School Gardens and Agricultural Projects**: Students can engage in gardening, farming, or horticulture activities, learning about agriculture, sustainability, and nutrition while developing a connection with nature.
- **Outdoor Adventure Activities**: Activities like camping, hiking, or orienteering promote teamwork, resilience, and leadership, while also providing opportunities for experiential learning in natural environments.

9. Student-Led Initiatives and Projects

- **Student Councils and Leadership Programs**: Involvement in student government or leadership programs allows students to develop leadership, organizational, and decision-making skills, while representing their peers and organizing school events.
- **Social Media and Digital Campaigns**: Encouraging students to create and run social media campaigns on issues they care about helps them develop digital literacy, communication, and advocacy skills.

10. Cross-Curricular Projects

- **Interdisciplinary Projects**: Projects that integrate multiple subjects, such as creating a documentary that involves research, writing, filming, and editing, help students make connections between different areas of learning and apply their knowledge in complex ways.
- **Thematic Learning**: Organizing activities around a central theme, such as "sustainability" or "innovation," encourages students to explore the theme from various perspectives and disciplines, fostering a deeper understanding of interconnected concepts.

Benefits of Non-Formal Activities in CBC

1. **Practical Application**:
    - These activities allow students to apply classroom learning in real-world situations, reinforcing their understanding and helping them see the relevance of their education.
2. **Development of Soft Skills**:
    - Non-formal activities promote the development of critical soft skills, such as communication, teamwork, leadership, creativity, and problem-solving, which are essential for success in the 21st century.
3. **Increased Engagement and Motivation**:
    - By offering diverse and engaging experiences, non-formal activities increase student motivation and enthusiasm for learning, making education more enjoyable and meaningful.
4. **Holistic Development**:
    - These activities contribute to the overall development of students, addressing not only academic competencies but also physical, social, emotional, and ethical growth.
5. **Community and Cultural Connection**:
    - Non-formal activities often involve community participation and cultural exploration, helping students build connections with their communities and develop a sense of identity and belonging.

In summary, non-formal activities are an integral part of a Competence-Based Curriculum, providing students with opportunities to develop and apply their competencies in diverse and meaningful ways. These activities complement formal education, supporting the holistic development of students and preparing them for the challenges and opportunities of the modern world

Topic 3

1. Differentiated learning

This is learning approach based on the fact that learners differ in their:

- Learning styles
- Needs and interests
- Strengths and abilities
- Learning pace
- Classroom activities should be adapted to meet these differences

Differentiated Learning in a **Competence-Based Curriculum (CBC)** is an instructional approach that tailors teaching strategies, learning activities, and assessments to meet the diverse needs, abilities, interests, and learning styles of individual students within the framework of developing specific competencies.

Key Concepts of Differentiated Learning in CBC

1. **Personalization of Learning**:
    - **Student-Centered Approach**: Differentiated learning places the student at the center of the educational process. It acknowledges that students learn in different ways and at different paces, so instruction is customized to suit their individual learning profiles.
    - **Flexible Learning Paths**: Teachers design various pathways for students to achieve the same competencies. This flexibility allows students to engage with the content in ways that align with their strengths, interests, and readiness levels.
2. **Diverse Teaching Strategies**:
    - **Varied Instructional Methods**: Teachers use a mix of instructional methods such as direct instruction, group work, project-based learning, and technology-enhanced activities to address different learning styles and intelligences.
    - **Scaffolded Learning**: Support is provided to students as needed, allowing those who require more help to receive it while enabling advanced learners to explore topics more deeply.
3. **Flexible Grouping**:
    - **Dynamic Grouping**: Students may be grouped and regrouped based on specific skills, interests, or learning needs. For instance, a student may work in one group for mathematics based on skill level and in another for a project based on interest in a particular topic.
    - **Collaborative Learning**: Differentiated learning often involves peer collaboration where students can learn from each other, offering opportunities for social learning and developing interpersonal skills.
4. **Assessment for Learning**:
    - **Formative Assessments**: Continuous assessments are used to gauge students' progress, identify learning gaps, and adjust instruction accordingly. These assessments may take different forms, such as quizzes, peer reviews, or self-assessments.

- **Multiple Forms of Assessment**: Competencies are assessed using various methods, such as projects, presentations, written assignments, and practical tasks. This approach allows students to demonstrate their understanding and skills in ways that suit their individual strengths.

5. **Respect for Diversity**:
    - **Cultural and Linguistic Sensitivity**: Differentiated learning in a CBC respects the diverse cultural and linguistic backgrounds of students, incorporating relevant content and practices that reflect students' identities.
    - **Inclusion of Special Needs**: Differentiated strategies are crucial for including students with special needs, ensuring that they can achieve the same competencies as their peers through tailored instruction and support.

How Differentiated Learning Enhances a Competence-Based Curriculum

1. **Maximizes Student Potential**: By recognizing and addressing individual differences, differentiated learning helps all students reach their full potential, ensuring that each student can achieve the required competencies.
2. **Engagement and Motivation**: When students engage with material in ways that align with their strengths and interests, they are more motivated and invested in their learning.
3. **Equity in Education**: Differentiated learning promotes equity by providing each student with the necessary resources and support to succeed, rather than applying a one-size-fits-all approach.
4. **Lifelong Learning Skills**: By catering to different learning styles and encouraging self-directed learning, differentiated instruction helps students develop skills that are essential for lifelong learning, such as critical thinking, problem-solving, and adaptability.

In summary, differentiated learning in a Competence-Based Curriculum ensures that education is adaptive to the needs of all students, allowing them to develop the necessary competencies in a way that is meaningful, relevant, and effective for them.

2. **Multiple Intelligences** in a **Competence-Based Curriculum** refers to the integration of Howard Gardner's theory of Multiple Intelligences into educational practices that focus on developing specific competencies in students. Here's a breakdown:

Multiple Intelligences Theory

Howard Gardner's theory of Multiple Intelligences, proposed in 1983, suggests that intelligence is not a single general ability but a combination of different kinds of intelligences. Gardner identified at least eight distinct intelligences:

1. **Linguistic Intelligence**: Sensitivity to spoken and written language.
2. **Logical-Mathematical Intelligence**: Ability to analyze problems logically and carry out mathematical operations.
3. **Spatial Intelligence**: Ability to recognize and manipulate patterns of space.
4. **Bodily-Kinesthetic Intelligence**: Using one’s whole body or parts of the body to solve problems or create products.
5. **Musical Intelligence**: Skills in performance, composition, and appreciation of musical patterns.
6. **Interpersonal Intelligence**: Capacity to understand and interact effectively with others.

7. **Intrapersonal Intelligence**: Capacity to understand oneself, including one's emotions and motivations.
8. **Naturalistic Intelligence**: Ability to recognize and categorize plants, animals, and other objects in nature.

Integration of Multiple Intelligences in CBC

In a Competence-Based Curriculum, the theory of Multiple Intelligences can be applied to ensure that the curriculum addresses the diverse ways in which students learn and demonstrate their competencies. This approach:

1. **Differentiated Instruction**: Teachers design lessons and activities that cater to different intelligences, ensuring that all students have the opportunity to develop their unique strengths.
2. **Personalized Learning Paths**: Students can pursue learning in ways that align with their dominant intelligences, making education more relevant and engaging.
3. **Assessment Flexibility**: Competence is assessed through various methods that allow students to demonstrate their understanding and skills in ways that best match their intelligences (e.g., oral presentations for linguistic intelligence, practical projects for bodily-kinesthetic intelligence).
4. **Holistic Development**: The curriculum fosters the overall development of the learner, recognizing that intelligence is multi-faceted and that different students will excel in different areas.

By integrating Multiple Intelligences into a Competence-Based Curriculum, educators can create a more inclusive and effective learning environment that respects and nurtures the unique abilities of each student.

Topic 4
Competence Based pedagogy/21st century teaching and learning methodologies

Competence-Based Pedagogy and **21st Century Teaching and Learning Methodologies** in a Competence-Based Curriculum (CBC) focus on equipping students with the skills, knowledge, and attitudes necessary to thrive in the modern world. These approaches emphasize active, student-centered learning, critical thinking, collaboration, and the practical application of knowledge in real-world contexts.

Competence-Based Pedagogy

Competence-Based Pedagogy is an instructional approach that prioritizes the development of specific competencies in students, going beyond traditional content delivery to focus on what students can do with their knowledge. This pedagogy is characterized by the following key elements:

1. **Student-Centered Learning**:
    - **Personalized Instruction**: Learning is tailored to the individual needs, interests, and abilities of each student. Teachers act as facilitators, guiding students as they take an active role in their own learning process.
    - **Flexible Learning Paths**: Students may have different pathways to achieve the same competencies, allowing for a more personalized and adaptive learning experience.
2. **Active Learning**:
    - **Experiential Learning**: Students engage in hands-on activities, simulations, and real-world projects that require them to apply their knowledge and skills in practical situations.
    - **Inquiry-Based Learning**: Encourages students to ask questions, conduct research, and explore topics deeply, fostering curiosity and critical thinking.
3. **Assessment for Learning**:
    - **Formative Assessment**: Continuous assessment is used to monitor student progress, provide feedback, and adjust instruction as needed. This approach helps ensure that students are developing the required competencies at their own pace.
    - **Competency-Based Assessment**: Assessments are designed to evaluate students' ability to apply what they have learned in practical, often interdisciplinary, contexts. This may include projects, presentations, portfolios, and performance tasks.
4. **Collaboration and Communication**:
    - **Peer Learning**: Students work in groups to solve problems, complete projects, and share knowledge, fostering collaboration and communication skills.
    - **Real-World Connections**: Learning experiences are connected to real-life scenarios, helping students understand the relevance and application of their competencies.
5. **Reflective Practice**:

- **Metacognition**: Students are encouraged to think about their own learning processes, set goals, and reflect on their progress, which promotes self-regulation and lifelong learning skills.

21st Century Teaching and Learning Methodologies

21st Century Teaching and Learning Methodologies are strategies that prepare students for the challenges of the modern world by focusing on key skills such as critical thinking, creativity, communication, and collaboration. These methodologies are closely aligned with competence-based pedagogy and include:

1. **Project-Based Learning (PBL)**:
    - **Student-Driven Projects**: Students engage in extended, interdisciplinary projects that require them to investigate and respond to complex questions or challenges. PBL develops critical thinking, problem-solving, and collaboration skills.
    - **Real-World Relevance**: Projects are often linked to real-world issues, making learning more meaningful and motivating for students.
2. **Problem-Based Learning (PrBL)**:
    - **Solving Complex Problems**: Students are presented with real-world problems and must work collaboratively to find solutions. This method emphasizes critical thinking, creativity, and application of knowledge in authentic contexts.
3. **Flipped Classroom**:
    - **Inverted Instruction**: Traditional learning models are reversed, with students first engaging with new content outside of class (e.g., through videos, readings), and then applying that knowledge in class through discussions, problem-solving activities, and projects.
    - **Active In-Class Learning**: Class time is dedicated to higher-order thinking activities, allowing for more personalized support from the teacher.
4. **Blended Learning**:
    - **Combining Online and Face-to-Face Learning**: Students experience a mix of digital and in-person learning activities. Blended learning allows for greater flexibility, self-paced learning, and the use of technology to enhance learning experiences.
5. **Collaborative Learning**:
    - **Group Work and Peer Learning**: Students work together in groups to complete tasks, solve problems, or create projects. This approach emphasizes teamwork, communication, and the ability to work effectively with others.
6. **Gamification**:
    - **Game-Based Learning**: Elements of game design, such as points, levels, and challenges, are incorporated into the learning process to increase student engagement and motivation.
    - **Interactive and Fun**: Learning becomes more interactive and enjoyable, encouraging students to take risks, experiment, and persist in the face of challenges.
7. **Design Thinking**:

- **Creative Problem Solving**: Students engage in a structured process of empathizing with users, defining problems, ideating solutions, prototyping, and testing. Design thinking fosters innovation, creativity, and practical problem-solving skills.

8. **STEAM Education**:
    - **Integration of Science, Technology, Engineering, Arts, and Mathematics**: STEAM education emphasizes the integration of these disciplines to develop students' ability to think critically, solve complex problems, and innovate.
    - **Interdisciplinary Learning**: Students work on projects that combine elements from multiple disciplines, helping them make connections across subjects and apply their learning in comprehensive ways.

Importance of Competence-Based Pedagogy and 21st Century Methodologies

1. **Preparation for the Future**:
    - These approaches equip students with the skills necessary for success in the 21st century, including adaptability, problem-solving, digital literacy, and collaboration.
2. **Engagement and Motivation**:
    - By making learning relevant, interactive, and student-centered, these methodologies increase student engagement and motivation, leading to deeper and more meaningful learning experiences.
3. **Development of Lifelong Learners**:
    - Students develop the ability to learn independently, think critically, and apply their knowledge in diverse contexts, which are essential skills for lifelong learning.
4. **Inclusive Education**:
    - Competence-based pedagogy and 21st-century methodologies support diverse learning needs, allowing for differentiation and personalized learning experiences that cater to all students.
5. **Focus on Real-World Application**:
    - These methods emphasize the practical application of knowledge and skills, ensuring that students are prepared to meet real-world challenges and contribute meaningfully to society.

In summary, Competence-Based Pedagogy and 21st Century Teaching and Learning Methodologies in a Competence-Based Curriculum focus on developing essential skills and competencies through active, student-centered, and relevant learning experiences. These approaches are designed to prepare students for the complexities of the modern world, fostering their ability to think critically, collaborate effectively, and continue learning throughout their lives.

Topic 5

Suggested learning and teaching resources

In a Competence-Based Curriculum (CBC), the selection of learning and teaching resources is crucial to effectively support the development of competencies in students. These resources should be diverse, engaging, and adaptable to different learning styles, allowing students to apply their knowledge in real-world contexts. Below is a list of suggested learning and teaching resources that can be used in CBC.

1. Digital Resources

- **Educational Software and Apps**: Tools like Khan Academy, Google Classroom, and Edmodo can provide interactive lessons, assessments, and collaborative opportunities.
- **Multimedia Content**: Videos, animations, and simulations from platforms like YouTube, TED-Ed, and National Geographic offer visual and auditory learning experiences that can help explain complex concepts.
- **Online Libraries and Databases**: Access to e-books, journals, and research databases such as JSTOR, Project Gutenberg, and Britannica Online can support research and deeper exploration of topics.
- **Learning Management Systems (LMS)**: Platforms like Moodle, Blackboard, and Canvas allow for the organization of courses, tracking of progress, and facilitation of online discussions.

2. Print Resources

- **Textbooks and Workbooks**: Well-designed textbooks aligned with the CBC can provide foundational knowledge and structured learning activities. Workbooks offer practice exercises and activities for skill development.
- **Reference Books**: Dictionaries, encyclopedias, and atlases can help students develop research skills and deepen their understanding of various subjects.
- **Educational Magazines and Journals**: Publications like "National Geographic Kids," "TIME for Kids," and subject-specific journals can provide up-to-date information and articles that are relevant to the curriculum.

3. Manipulatives and Hands-On Materials

- **Mathematics Manipulatives**: Tools such as base-ten blocks, fraction tiles, and geometric shapes can help students visualize and understand mathematical concepts.
- **Science Kits**: Experiment kits for biology, chemistry, and physics enable students to conduct hands-on experiments and explore scientific principles.
- **Art Supplies**: Materials like paints, clay, and craft supplies encourage creativity and help students express their ideas through visual arts.
- **Engineering and Robotics Kits**: Kits like LEGO Education, Arduino, and Raspberry Pi allow students to engage in STEM activities, building and programming models that demonstrate engineering and technological concepts.

4. Real-World Resources

- **Field Trips**: Visits to museums, science centers, historical sites, nature reserves, and local businesses provide experiential learning opportunities that connect classroom knowledge to the real world.
- **Guest Speakers and Experts**: Inviting professionals from various fields to speak with students or participate in discussions can provide insights into real-world applications of their learning.
- **Community Resources**: Collaborations with local organizations, government agencies, and community groups can offer practical experiences, such as service-learning projects or internships.
- **Case Studies and Real-Life Scenarios**: Using case studies or real-life scenarios related to current events, industry practices, or social issues can help students apply their learning in practical contexts.

5. Collaborative and Interactive Resources

- **Group Work Tools**: Resources like collaborative whiteboards (e.g., Jamboard), project management apps (e.g., Trello), and shared documents (e.g., Google Docs) facilitate teamwork and group projects.
- **Peer Learning Resources**: Activities like peer review sessions, study groups, and collaborative problem-solving exercises encourage students to learn from each other and develop social skills.
- **Role-Playing and Simulations**: Interactive simulations, role-playing games, and mock trials can help students develop empathy, decision-making skills, and a deeper understanding of complex issues.

6. Teacher Resources

- **Professional Development Materials**: Books, webinars, and online courses on CBC implementation, differentiated instruction, and 21st-century teaching strategies can support teacher growth and effectiveness.
- **Curriculum Guides and Lesson Plans**: Access to curriculum frameworks, lesson plan templates, and instructional guides aligned with the CBC helps teachers design effective learning experiences.
- **Assessment Tools**: Rubrics, checklists, and formative assessment strategies tailored to competency-based evaluation can assist teachers in measuring student progress and providing meaningful feedback.

7. Cultural and Linguistic Resources

- **Culturally Relevant Materials**: Books, stories, music, and art that reflect the cultural backgrounds and languages of students help to create an inclusive learning environment and promote cultural understanding.
- **Language Learning Resources**: Tools like language apps (e.g., Duolingo), bilingual dictionaries, and language exchange programs support students in developing linguistic competencies, especially in multilingual settings.

8. Interactive and Creative Resources

- **Games and Puzzles**: Educational games, both digital and physical, and puzzles that challenge students' problem-solving and critical thinking abilities can make learning fun and engaging.
- **Creative Writing and Storytelling Tools**: Story starters, writing prompts, and digital storytelling apps (e.g., Story bird) encourage students to develop their communication skills through creative expression.

9. Assessment Resources

- **Portfolio Tools**: Platforms like Seesaw or physical portfolios where students can collect and reflect on their work, demonstrating their progress in competencies over time.
- **Rubrics and Checklists**: Standardized tools that help assess competencies consistently across different students and tasks, ensuring that all aspects of a competency are measured.

In summary, the suggested learning and teaching resources in a Competence-Based Curriculum should be diverse, interactive, and aligned with the competencies being developed. These resources support a wide range of learning experiences, from hands-on activities and real-world applications to digital learning and collaborative projects, ensuring that students can engage with the material in meaningful ways and develop the skills needed for success in the 21st century.

Topic 6
Introduction to inclusive Education

Introduction to Inclusive Education in a Competence-Based Curriculum (CBC)
Inclusive education is a fundamental approach within a Competence-Based Curriculum (CBC), designed to ensure that all learners, regardless of their abilities, backgrounds, or needs, have equitable access to quality education. The core idea behind inclusive education in a CBC is to create a learning environment where every student can succeed and develop competencies at their own pace and in a manner that suits their unique needs.

What is Inclusive Education?
Inclusive education refers to the practice of educating all students together in a way that accommodates their diverse needs. It is built on the belief that every child has the right to a quality education, and that schools should adapt to meet the needs of all students, rather than expecting students to adapt to the school. This approach emphasizes diversity, equity, and the removal of barriers to learning for all students, including those with disabilities, special educational needs, or those from marginalized or disadvantaged groups.

Principles of Inclusive Education in CBC

1. **Equity and Access**:
    - **Universal Access**: Inclusive education in CBC ensures that all students, regardless of their physical, intellectual, emotional, or social differences, have access to the curriculum and can participate fully in all learning activities.
    - **Equitable Opportunities**: Schools and educators provide equitable learning opportunities by adapting teaching methods, resources, and assessments to meet the diverse needs of students.
2. **Personalized Learning**:
    - **Differentiated Instruction**: Teachers use differentiated teaching strategies to cater to the diverse learning styles, interests, and abilities of all students, ensuring that each student can engage with and benefit from the curriculum.
    - **Individualized Support**: Some students may require individualized support, such as special education services, modified learning materials, or assistive technologies, to help them achieve their learning goals.
3. **Collaboration and Community**:
    - **Inclusive Classroom Environment**: An inclusive classroom is one where all students feel valued, respected, and part of the learning community. Teachers foster a supportive atmosphere where diversity is celebrated, and every student has the opportunity to contribute.
    - **Collaboration with Families and Communities**: Inclusive education involves close collaboration with families, caregivers, and community resources to support the learning and development of all students. This partnership ensures that students' needs are met both inside and outside the classroom.
4. **Focus on Strengths and Potential**:

- **Strength-Based Approach**: Inclusive education in CBC focuses on students' strengths, talents, and potential rather than their limitations. Educators identify and build on each student's unique abilities to help them achieve success.
- **High Expectations for All**: Teachers hold high expectations for all students, regardless of their starting points, and provide the necessary support to help them reach their full potential.

5. **Flexible and Adaptive Curriculum**:
    - **Competence-Based Learning**: CBC emphasizes the development of competencies—skills, knowledge, and attitudes—rather than merely covering content. This allows for a more flexible curriculum that can be adapted to meet the needs of diverse learners.
    - **Adaptation and Modification**: The curriculum can be adapted or modified to accommodate the needs of students with disabilities or special educational needs, ensuring that they can participate in and benefit from the learning process.

Strategies for Implementing Inclusive Education in CBC

1. **Differentiated Instruction**:
    - Teachers design lessons that cater to different learning styles and levels of readiness, using a variety of teaching methods, such as group work, visual aids, hands-on activities, and technology-assisted learning.
2. **Use of Assistive Technology**:
    - Incorporating assistive technologies, such as screen readers, speech-to-text software, and adaptive learning tools, can help students with disabilities access the curriculum and participate fully in classroom activities.
3. **Universal Design for Learning (UDL)**:
    - UDL is an educational framework that involves designing curricula that are accessible to all students from the outset, by providing multiple means of representation, expression, and engagement.
4. **Peer Support and Collaboration**:
    - Encouraging peer tutoring, cooperative learning groups, and collaborative projects helps students learn from one another and fosters a sense of community and belonging.
5. **Professional Development for Teachers**:
    - Ongoing training and professional development are essential for teachers to effectively implement inclusive practices. This includes training in differentiated instruction, classroom management, and the use of assistive technologies.
6. **Positive Behavioral Support**:
    - Implementing positive behavioral interventions and supports (PBIS) helps create a safe and supportive classroom environment where all students can thrive.

Benefits of Inclusive Education in CBC

1. **Enhanced Learning for All**:

    - Inclusive education benefits all students by fostering a culture of empathy, respect, and collaboration. It encourages the development of social skills, teamwork, and a deeper understanding of diversity.
2. **Improved Academic and Social Outcomes**:
    - Research shows that inclusive education can lead to better academic and social outcomes for students with and without disabilities, as they learn to appreciate different perspectives and support one another.
3. **Preparation for a Diverse Society**:
    - Inclusive education prepares students for life in a diverse society by teaching them to value differences, work collaboratively, and develop inclusive attitudes and behaviors.
4. **Reduction of Discrimination and Stigmatization**:
    - By educating students together, inclusive education reduces the stigma and discrimination often associated with special education and marginalized groups, promoting equality and social justice.

Challenges and Considerations

- **Resource Allocation**: Effective inclusive education requires adequate resources, including trained staff, assistive technologies, and appropriate learning materials. Schools may face challenges in securing these resources.
- **Teacher Preparedness**: Teachers need to be well-prepared and supported to implement inclusive practices. This may require ongoing professional development and collaboration with special education professionals.
- **Cultural and Attitudinal Barriers**: Overcoming societal attitudes and cultural norms that may resist the inclusion of students with disabilities or differences is essential for successful implementation.

Conclusion

Inclusive education in a Competence-Based Curriculum is about ensuring that every student, regardless of their individual needs and abilities, can access and benefit from education. It is grounded in the principles of equity, diversity, and respect for all learners. By embracing inclusive practices, schools can create a learning environment where all students are supported to achieve their full potential, preparing them for successful and fulfilling lives in a diverse and interconnected world.

Topic 7

Mainstreaming values (Value Based Education and PCIs)

In a Competence-Based Curriculum (CBC), mainstreaming values such as ethics, integrity, responsibility, and respect is essential to develop well-rounded individuals who can contribute positively to society. Value-Based Education (VBE) and the integration of Pertinent Contemporary Issues (PCIs) are key strategies for embedding these values into the curriculum, ensuring that students not only acquire academic knowledge and skills but also develop the moral and ethical compass necessary for personal and social development.

Value-Based Education (VBE) in CBC

Value-Based Education is an approach that integrates core values into the curriculum, teaching students to understand, appreciate, and apply these values in their daily lives. In a CBC, VBE goes beyond theoretical knowledge, focusing on the practical application of values in real-life situations, thereby fostering the development of character, ethical decision-making, and social responsibility.

Key Components of Value-Based Education

1. **Core Values Identification**:
    - Core values such as honesty, empathy, respect, responsibility, fairness, and integrity are identified and explicitly taught throughout the curriculum. These values are woven into every subject area and extracurricular activity.
2. **Role Modeling**:
    - Teachers and school staff play a crucial role in modeling the values they teach. By demonstrating these values in their interactions with students and colleagues, they set a positive example for students to follow.
3. **Integrated Curriculum**:
    - Values are integrated across all subjects, not taught in isolation. For example, in literature, students might explore themes of justice and morality; in science, discussions might focus on environmental ethics and responsible use of technology.
4. **Experiential Learning**:
    - Students engage in activities that allow them to practice values in real-world contexts, such as community service projects, peer mentoring, and collaborative group work.
5. **Reflective Practices**:
    - Students are encouraged to reflect on their actions, decisions, and learning experiences. This reflection helps them internalize values and understand their importance in everyday life.
6. **Assessment of Values**:

- The assessment of values is done through observation of behavior, self-assessment, and peer assessment, rather than traditional exams. This includes evaluating how students apply values in different situations.

Pertinent Contemporary Issues (PCIs) in CBC

Pertinent Contemporary Issues (PCIs) are current social, environmental, and global challenges that are relevant to students' lives and the broader society. Mainstreaming PCIs in a CBC ensures that education is relevant, responsive, and prepares students to engage with and address these issues effectively.

Key PCIs Typically Integrated into CBC

1. **Environmental Sustainability**:
   - Topics such as climate change, conservation, sustainable development, and responsible consumption are integrated into the curriculum, encouraging students to adopt sustainable practices and become stewards of the environment.
2. **Social Justice and Equity**:
   - Issues like gender equality, human rights, anti-discrimination, and social inclusion are explored, helping students understand and advocate for a fair and just society.
3. **Health and Well-being**:
   - Students learn about physical and mental health, nutrition, fitness, and well-being, as well as issues like substance abuse, HIV/AIDS awareness, and healthy relationships.
4. **Global Citizenship**:
   - The curriculum includes discussions on global interdependence, cultural diversity, peace, and conflict resolution, preparing students to be informed and active global citizens.
5. **Digital Literacy and Cyber Ethics**:
   - With the increasing role of technology, students are taught about digital citizenship, online safety, cyberbullying, and the ethical use of digital resources.
6. **Economic Literacy and Entrepreneurship**:
   - Lessons on financial literacy, entrepreneurship, and ethical business practices equip students with the skills and values needed to navigate and contribute to the economy responsibly.

Strategies for Mainstreaming Values and PCIs in CBC

1. **Curriculum Integration**:
   - Values and PCIs are embedded in all subject areas. For example, in social studies, students might analyze historical events through the lens of social justice; in science, they might study the impact of human activity on the environment.
2. **Project-Based Learning**:
   - Students work on projects that address real-world issues, such as creating a sustainability plan for their school or developing a community outreach program. These projects help students apply their learning and values in meaningful ways.

3. **Discussion and Debate**:
    - Class discussions, debates, and ethical dilemmas are used to explore values and PCIs. These activities encourage critical thinking, empathy, and the development of informed opinions.
4. **Service Learning**:
    - Service learning combines community service with academic learning, allowing students to apply their knowledge and values in service to others. For example, students might organize a campaign to raise awareness about a local environmental issue.
5. **Interdisciplinary Approach**:
    - An interdisciplinary approach helps connect values and PCIs across different subjects, showing students how these issues intersect in the real world. For example, a unit on water conservation might include science (water cycle), social studies (water rights), and art (water-themed projects).
6. **School Culture and Policies**:
    - The school culture, policies, and practices should reflect the values being taught. This includes promoting respect, inclusivity, and equity within the school community and addressing any behaviors that contradict these values.

Benefits of Mainstreaming Values and PCIs in CBC

1. **Holistic Development**:
    - By integrating values and PCIs, the CBC promotes the holistic development of students, equipping them with the knowledge, skills, and attitudes necessary to navigate and contribute positively to society.
2. **Enhanced Critical Thinking and Problem-Solving**:
    - Engaging with PCIs encourages students to think critically about complex issues, consider different perspectives, and develop solutions to real-world problems.
3. **Preparation for Real-Life Challenges**:
    - Students learn to apply their competencies and values in real-world contexts, preparing them to face contemporary challenges and make informed, ethical decisions.
4. **Fostering Social Responsibility**:
    - Value-based education and the integration of PCIs instill a sense of social responsibility in students, motivating them to contribute to the well-being of their communities and the world.
5. **Promotion of Ethical Behavior**:
    - By emphasizing values such as integrity, empathy, and respect, the CBC helps cultivate ethical behavior in students, both in their personal lives and in their future professional careers.

Challenges and Considerations

- **Cultural Sensitivity**: It is important to approach values education with cultural sensitivity, recognizing and respecting the diverse backgrounds and beliefs of students and their communities.

- **Teacher Preparedness**: Teachers need adequate training and support to effectively integrate values and PCIs into their teaching. This includes understanding how to facilitate discussions on potentially sensitive or controversial topics.
- **Assessment**: Measuring the development of values and competencies related to PCIs can be challenging. Schools may need to develop innovative assessment methods that capture these aspects of learning.

Conclusion

Mainstreaming values through Value-Based Education and integrating Pertinent Contemporary Issues in a Competence-Based Curriculum is essential for preparing students to be ethical, responsible, and informed citizens. By embedding these elements into the curriculum, educators can ensure that students not only acquire academic knowledge but also develop the moral and ethical foundations necessary to navigate the complexities of the modern world and contribute positively to society. This approach fosters holistic development, critical thinking, and social responsibility, aligning education with the broader goals of personal growth and societal well-being.

Topic 8

Introduction to Community Service Learning

Community Service Learning (CSL) is an educational approach that combines academic learning with meaningful community service. In a Competence-Based Curriculum (CBC), CSL is a powerful tool for fostering practical, real-world skills and competencies while addressing community needs. It provides students with opportunities to apply their knowledge and skills in authentic contexts, promoting personal growth, civic responsibility, and social impact.

What is Community Service Learning?

Community Service Learning (CSL) is a pedagogical strategy that integrates community service with academic instruction. It involves students participating in service activities that benefit the community, while reflecting on their experiences and linking them to academic learning objectives. CSL is designed to enrich students' learning experiences, deepen their understanding of course material, and enhance their sense of social responsibility.

Principles of Community Service Learning in CBC

1. **Integration of Service and Learning**:
    - CSL connects community service activities with academic learning goals. The service projects are designed to reinforce and apply the competencies and knowledge students are acquiring through their coursework.
2. **Reciprocal Benefit**:
    - The CSL approach benefits both students and the community. Students gain practical experience and develop skills, while the community receives valuable services and support.
3. **Reflection and Analysis**:
    - Reflection is a key component of CSL. Students engage in structured reflection activities to analyze their experiences, understand the impact of their service, and connect their learning to real-world issues.
4. **Student Choice and Ownership**:
    - Students have a voice in selecting and designing their service projects. This fosters a sense of ownership and personal investment in the learning process.
5. **Collaboration and Partnership**:
    - CSL involves collaboration with community organizations and stakeholders. Partnerships between schools and community groups are essential for identifying needs and coordinating service projects.

Benefits of Community Service Learning in CBC

1. **Enhanced Learning**:
    - CSL provides students with practical experiences that reinforce and contextualize academic learning. Applying classroom knowledge to real-world problems helps students understand and retain concepts more effectively.
2. **Development of Competencies**:
    - Through CSL, students develop a range of competencies, including critical thinking, problem-solving, teamwork, communication, and leadership skills.

These competencies are essential for success in both academic and professional contexts.

3. **Increased Civic Engagement**:
    - CSL fosters a sense of civic responsibility and social awareness. Students learn about community needs, social issues, and the importance of contributing to the common good.
4. **Personal Growth**:
    - Participating in CSL activities helps students build self-confidence, empathy, and resilience. They gain insights into their own values and interests while developing a deeper understanding of diverse perspectives.
5. **Strengthened Community Ties**:
    - CSL creates strong connections between schools and communities. By addressing local needs, students contribute to community development and build positive relationships with community members and organizations.

Strategies for Implementing Community Service Learning in CBC

1. **Curriculum Integration**:
    - Design CSL projects that align with the learning objectives of the CBC. Ensure that the service activities are connected to academic content and competencies, allowing students to apply their learning in practical ways.
2. **Partnership Development**:
    - Establish partnerships with local community organizations, businesses, and service providers. Collaborate with these partners to identify community needs and design meaningful service projects.
3. **Project Planning and Execution**:
    - Involve students in the planning and execution of their service projects. Encourage them to identify community needs, develop project plans, and implement their ideas, fostering a sense of ownership and responsibility.
4. **Structured Reflection**:
    - Incorporate regular reflection activities throughout the CSL process. Use journals, essays, group discussions, and presentations to help students analyze their experiences, connect their service to academic learning, and articulate their personal growth.
5. **Assessment and Evaluation**:
    - Develop assessment tools to evaluate both the academic and service components of CSL. Assess students' learning outcomes, the impact of their service on the community, and their personal development.
6. **Support and Guidance**:
    - Provide support and guidance to students throughout their CSL experiences. Offer resources, mentorship, and feedback to help them navigate their projects and reflect on their learning.
7. **Celebrating Achievements**:

    - Recognize and celebrate students' contributions and achievements in CSL. Acknowledge their efforts through awards, presentations, or community events to reinforce the value of their service.

Examples of Community Service Learning Projects

1. **Environmental Conservation**:
    - Students work on projects such as tree planting, recycling programs, or environmental education campaigns, applying their knowledge of science and sustainability.
2. **Tutoring and Mentoring**:
    - Students provide academic support and mentorship to younger students or peers, using their skills and knowledge to help others succeed.
3. **Health and Wellness Initiatives**:
    - Students participate in health promotion activities, such as organizing wellness workshops, conducting health screenings, or raising awareness about healthy living.
4. **Community Development**:
    - Students engage in projects that address local community needs, such as organizing food drives, building community gardens, or supporting local shelters.
5. **Cultural and Arts Projects**:
    - Students collaborate on cultural events, art exhibitions, or performances that celebrate diversity and promote cultural understanding.

Challenges and Considerations

- **Resource Availability**: Ensuring that there are adequate resources and support for CSL projects can be a challenge. Schools need to allocate time, funding, and personnel to support successful implementation.
- **Student Engagement**: Keeping students motivated and engaged in CSL activities requires careful planning and meaningful projects that align with their interests and skills.
- **Assessment**: Developing effective assessment methods for CSL can be complex, as it involves evaluating both academic and service components. Schools need to design comprehensive assessment tools that capture the impact and learning outcomes.

Conclusion

Community Service Learning in a Competence-Based Curriculum is a valuable approach that integrates academic learning with community engagement. By participating in CSL, students develop essential competencies, gain practical experience, and contribute positively to their communities. CSL enhances the educational experience by making learning relevant and impactful, fostering personal growth, and promoting civic responsibility. Through well-designed CSL projects, students can bridge the gap between classroom knowledge and real-world application, preparing them to be informed, compassionate, and active members of society.

Topic 9

Competence Based Assessment Framework and Assessment tools

Overview of CBC and CBA

By the end of the session, the participant should be able to:

a) explain the meaning of CBC and CBA
b) explain the linkage between CBC and CBA
c) discuss the purpose of assessment
d) explain the CBA Framework
e) discuss the Paradigm shifts in CBA
f) discus types of assessment

Basic Education Curriculum Framework (Age-based)

	BECF	Grades	Age	Years
a)	Early years Education - Pre-primary	PP1 - PP2	4 - 5	2
b)	Early years Education - Lower Primary	1-3	6-8	3
c)	Middle school Education-Upper Primary	4-6	9-11	3
d)	Middle School Education-Junior Secondary	7-9	12-14	3
e)	Senior School	10-12	15-17	3
f)	University	-	18-20	3

MAJOR SHIFTS

NEW SYSTEM

Focuses on the nurturing of a holistic individual in character and skill in line with the tenets of the 2010 Constitution of Kenya and Kenya Vision 2030.

PREVIOUS SYSTEM

From 7-4-2-3 in 1963, to 8-4-4 in 1985 – over time the system became academic and examination-oriented resulting in three key problems**:**

1. Disconnect between school content and the world of work. Most learners exit school without skills for work and life hence increased unemployment.
2. Deterioration in character/value system. Laying emphasis on the entrenchment of core competencies and values seeks to address this.
3. Disconnect with the 21st century needs. The system focuses on the development of a creative, multi-skilled thinkers who will fit in the rapidly changing world.

Shift in Assessment

Object assessment

1. Assessment of learning (summative assessment)
2. Assessment of knowledge acquired through rote learning
3. Norm referenced
4. Encourages competition
5. Rigid (prescribed duration)
6. Teacher and national assessment

7. Quantitative reporting based on position and marks
8. Memorization and reproduction
9. Assessment of lower order skills
10. Authentic assessment tasks

Competency Based Assessment

1. Assessment for and as learning (formative assessment)
2. Assessment of competencies development
3. Criterion referenced
4. Encourages collaboration and cooperation
5. Flexible based on the pace of the learner
6. Includes self and peer assessment
7. Both quantitative and qualitative (description of the competency demonstrated)
8. Demonstration of creativity and talents (Performance Based)
9. Assessment of higher order skill
10. Authentic assessment tasks

Meaning of CBC

What is a competency?

Is the ability to apply appropriate knowledge, skills, values and attitude to successfully perform a real-life task.

Competency Based Curriculum (CBC)

An education programme focused on the learner's ability to apply the knowledge, skills, values and attitude to successfully perform a real-life tasks.

Meaning of Competency Based Assessment (CBA)

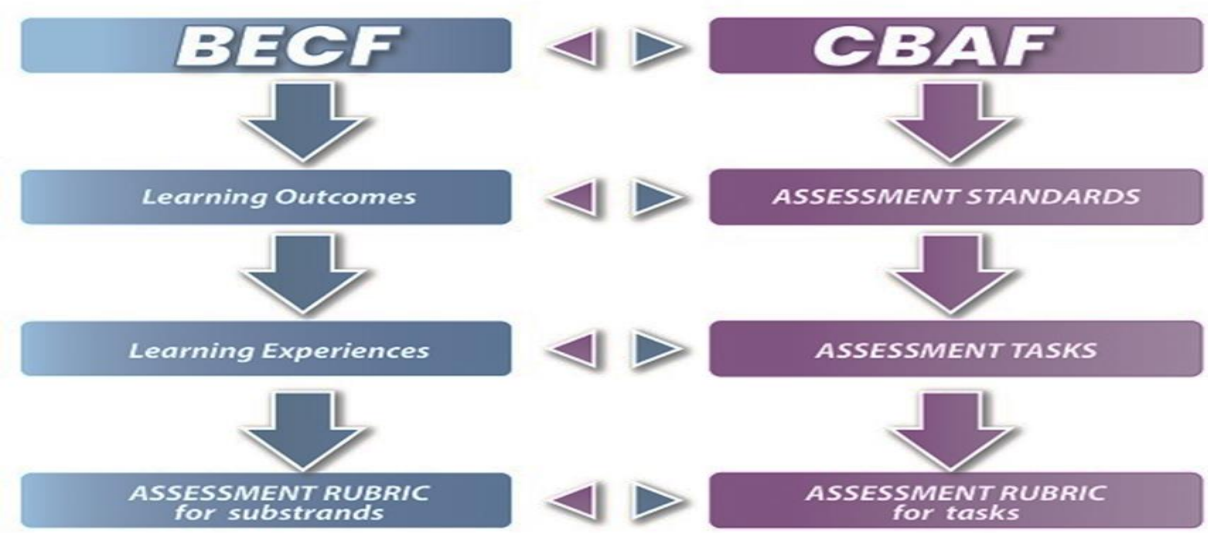
- Is the process of determining the learner's ability to apply a set of knowledge, skills, values and attitudes to successfully perform a real-life task?
- Is the process of gathering and interpreting information on what the learner knows and can do against the specified learning outcomes?
- This is done using a variety of assessment tools such as written tests, checklists, questionnaires, observation schedules, journals, portfolios and rubrics

Objectives of Competency Based Assessment Framework

- Evidence for *Accountability*
- Identifying and Nurturing *Learner Potential*
- Determine acquisition of competencies and values
- Improving *Learning, Instruction & Assessment*
- Measuring Learner Participation in *CSL & Acquisition of Values*
- Feedback to *Stakeholders*
- For certification after Grade 12

Linkage between various components of CBC and CBA

Acronyms used in the Competency Based Assessment Framework Acronyms used in the Competency



Acronyms used in the Competency Based Assessment Framework

CA – Classroom Assessment

SBA – School Based Assessment

KEYA – Kenya Early Years Assessment

SYR – School Year Report

KPSEA – Kenya Primary School Education Assessment

KJSEA – Kenya Junior School Education Assessment

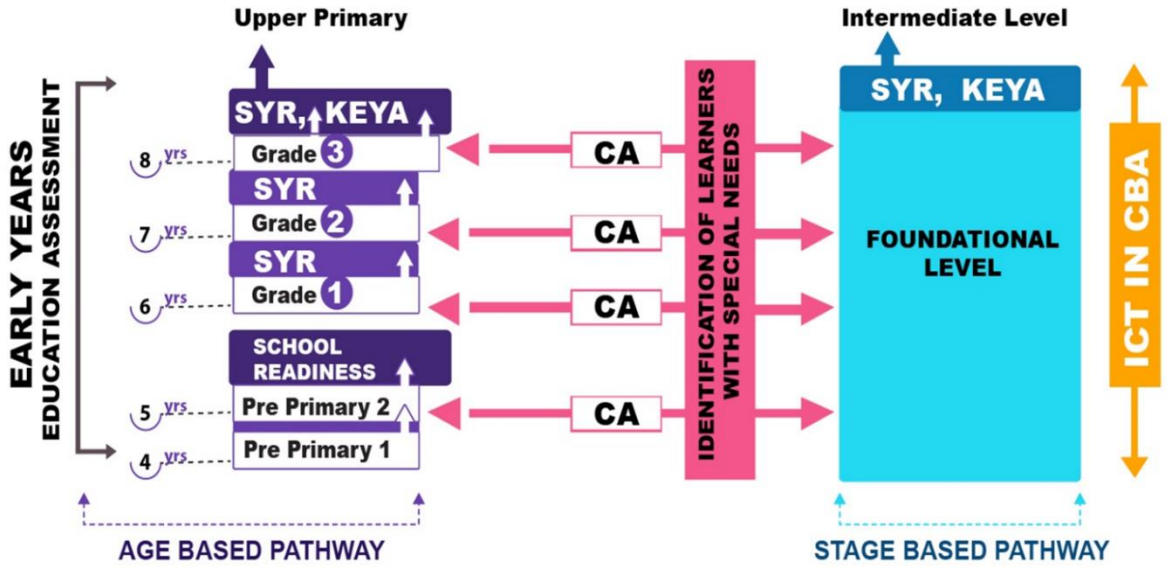
KCBE – Kenya Certificate of Basic Education

LEP – Learners Exit Profile

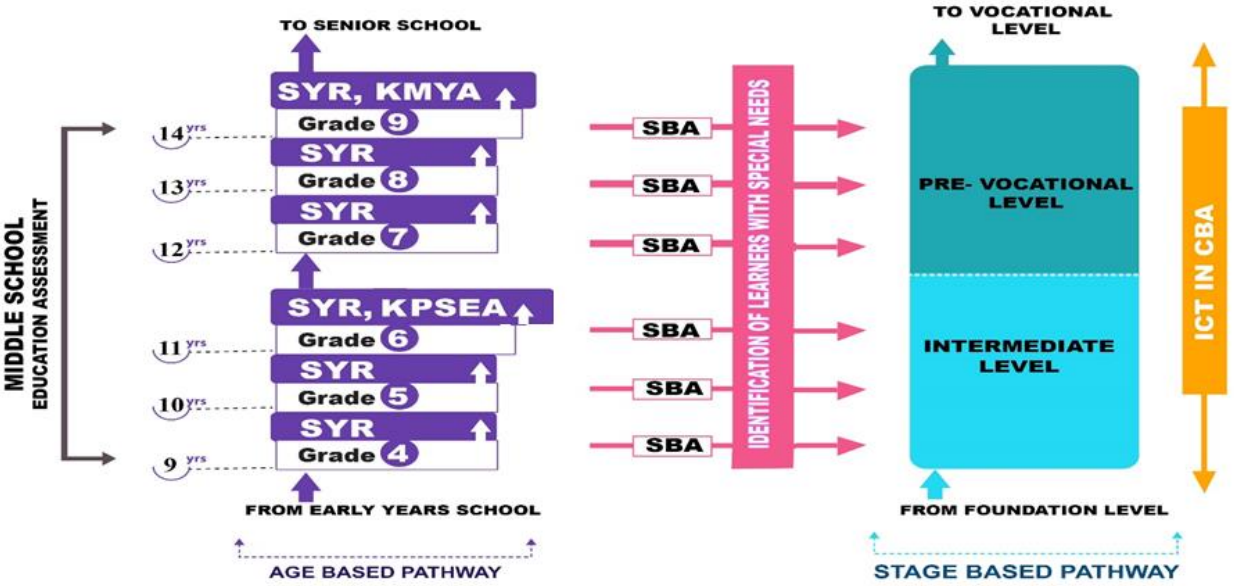
Levels of Competency Based Assessment
CBA is divided into three levels

a) Early Years Education Assessment
b) Middle School Education Assessment
c) Senior School Education Assessment

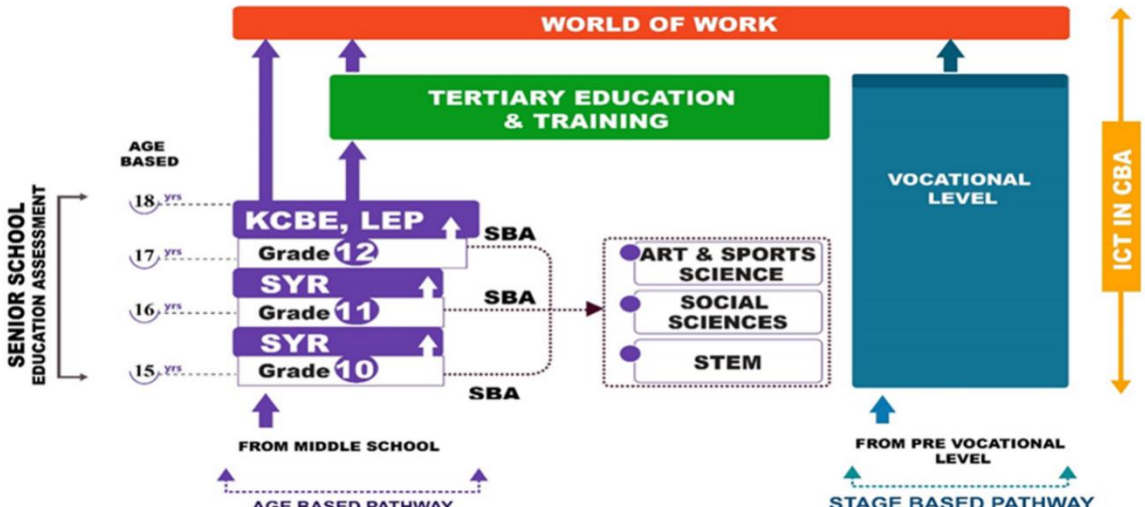
STRUCTURE OF EARLY YEARS EDUCATION ASSESSMENT



STRUCTURE OF MIDDLE SCHOOL EDUCATION ASSESSMENT



STRUCTURE OF SENIOR SCHOOL EDUCATION ASSESSMENT



PRINCIPLES GUIDING ASSESSMENT

Validity

accurately measures what it intends to measure

Reliability

consistency in producing identical results

Fairness

equality, equity and justice

Flexibility

Practicability

- economy of time, effort and cost

Authenticity

- assessment tasks relate to real-life experiences

Sufficiency

- sufficient evidence to reach an assessment decision

Timely Feedback

- immediate

Collaboration

- involves self, peer and teacher assessment

Currency

- recognition of current competencies

TYPES OF ASSESSMENT

- **Formative Assessment**
  - Assessment *for* learning – designed and administered during the learning process
  - Assessment *as* learning – learners evaluate their work against learning outcomes
  - For example, Classroom Assessment

    - And school Based Assessment
- **Summative Assessment**
    - Assessment *of* learning – undertaken at the end of a learning period
    - For example, National Assessment

Theories of learning and Assessment

- **Cognitive Development**
    - Piaget's Cognitive Development Theory
    - Bruner's Cognitive Development Theory
    - Howard Gardner's Multiple Intelligence Theory
- **Social Constructivism**
    - Dewey's Social Constructivism amplified by
        - Elliot
        - Jonassen
        - Honebein
    - Vygotsky's Sociocultural Theory
- **Behaviourism**
    - Ivan Pavlov
    - Edward Thorndike
    - John B. Watson
    - B. F. Skinner

EDUCATIONAL STAKEHOLDERS IN THE CBAF

Competence Based Assessment Framework

A Competence-Based Assessment Framework is integral to a Competence-Based Curriculum (CBC). It focuses on evaluating students based on their ability to demonstrate specific competencies—skills, knowledge, and attitudes—required to perform effectively in real-world contexts. This framework ensures that assessments are aligned with the goals of the CBC, providing a clear and meaningful measure of students' proficiency and readiness.

Key Elements of a Competence-Based Assessment Framework

1. **Clear Competency Statements**:
    - **Definition**: Competency statements outline what students are expected to know, do, and value by the end of a learning period. These statements should be specific, measurable, and aligned with the learning objectives of the curriculum.
    - **Examples**: "Apply problem-solving techniques to real-world scenarios," "Demonstrate effective communication skills in group settings," or "Understand and analyze ethical issues in professional contexts."
2. **Assessment Criteria**:
    - **Criteria Development**: Develop clear criteria for evaluating students' performance against each competency. These criteria should outline the expectations for different levels of proficiency and provide benchmarks for assessment.
    - **Levels of Achievement**: Define various levels of achievement (e.g., basic, proficient, advanced) to differentiate between varying degrees of competency demonstrated by students.
3. **Formative and Summative Assessments**:
    - **Formative Assessment**: Ongoing assessments that provide feedback to students during the learning process. These assessments help identify strengths and areas for improvement and guide instructional adjustments.
    - **Summative Assessment**: Evaluations at the end of a learning period to determine whether students have achieved the required competencies. Summative assessments typically include final exams, projects, presentations, or practical demonstrations.
4. **Authentic Assessment Methods**:
    - **Real-World Tasks**: Use assessment methods that simulate real-world tasks and scenarios relevant to the competencies being evaluated. This can include case studies, role-plays, simulations, and practical projects.
    - **Performance-Based Assessments**: Evaluate students' ability to perform tasks or solve problems in authentic contexts, such as conducting experiments, creating presentations, or managing projects.
5. **Holistic Assessment**:
    - **Comprehensive Evaluation**: Assess students' overall performance by considering multiple aspects of competency, including knowledge, skills, and attitudes. This approach ensures a well-rounded evaluation of students' abilities.

- **Integration of Multiple Sources**: Use various assessment sources, such as observations, self-assessments, peer assessments, and teacher evaluations, to provide a comprehensive view of students' competencies.

6. **Student Involvement**:
   - **Self-Assessment**: Encourage students to reflect on their own learning and evaluate their progress toward achieving competencies. Self-assessment helps students develop self-awareness and take ownership of their learning.
   - **Peer Assessment**: Facilitate peer evaluations where students assess each other's performance based on established criteria. Peer assessment promotes collaborative learning and provides diverse perspectives on competency development.
7. **Feedback and Reflection**:
   - **Constructive Feedback**: Provide timely and constructive feedback to students, highlighting strengths and areas for improvement. Feedback should be specific, actionable, and focused on helping students enhance their competencies.
   - **Reflective Practice**: Encourage students to reflect on their performance and learning experiences. Reflection helps students understand their progress, set goals, and identify strategies for further development.
8. **Alignment with Learning Objectives**:
   - **Consistency**: Ensure that assessment methods and criteria align with the learning objectives of the CBC. This alignment guarantees that assessments accurately measure the competencies intended to be developed.
   - **Relevance**: Design assessments that are relevant to the real-world applications of the competencies, ensuring that students' learning experiences are meaningful and applicable.

Implementing the Competence-Based Assessment Framework

1. **Design Assessment Tools**:
   - Create rubrics, checklists, and scoring guides that outline the criteria and levels of achievement for each competency. These tools should be clear, detailed, and aligned with the competency statements.
2. **Integrate Assessments into the Curriculum**:
   - Embed formative and summative assessments throughout the curriculum to continuously monitor and evaluate students' progress. Ensure that assessments are integrated with instructional activities and learning experiences.
3. **Provide Professional Development**:
   - Train teachers and assessors on the principles and practices of competence-based assessment. Professional development should focus on designing effective assessments, providing feedback, and using assessment data to inform instruction.
4. **Monitor and Evaluate the Framework**:
   - Regularly review and evaluate the effectiveness of the assessment framework. Gather feedback from students, teachers, and stakeholders to make necessary adjustments and improvements.

Benefits of a Competence-Based Assessment Framework

1. **Focused on Real-World Skills**:
    - By assessing competencies rather than just knowledge, the framework ensures that students develop and demonstrate skills relevant to real-world contexts and professional success.
2. **Encourages Deep Learning**:
    - Competence-based assessment promotes deeper learning by emphasizing the application of knowledge and skills rather than rote memorization. This approach supports critical thinking and problem-solving abilities.
3. **Provides Meaningful Feedback**:
    - Clear assessment criteria and authentic tasks provide students with meaningful feedback on their performance, helping them understand their strengths and areas for growth.
4. **Supports Personalized Learning**:
    - The framework allows for personalized assessment based on individual students' needs and progress. It provides opportunities for differentiated instruction and targeted support.
5. **Enhances Student Motivation**:
    - By demonstrating the relevance of competencies to real-world situations, students are more likely to be motivated and engaged in their learning. The focus on practical skills and application can increase their sense of purpose and achievement.

Challenges and Considerations

- **Design Complexity**: Developing a comprehensive competence-based assessment framework can be complex and time-consuming. It requires careful planning, collaboration, and alignment with curriculum goals.
- **Consistency in Evaluation**: Ensuring consistency and fairness in the assessment of competencies across different assessors and contexts can be challenging. Clear criteria and training are essential to maintain consistency.
- **Resource Requirements**: Implementing and maintaining the framework may require additional resources, such as assessment tools, professional development, and time for planning and evaluation.

Conclusion

A Competence-Based Assessment Framework is essential for effectively evaluating student progress in a Competence-Based Curriculum. By focusing on the demonstration of specific competencies, this framework ensures that assessments are aligned with real-world applications and provide meaningful feedback. Through clear criteria, authentic tasks, and a holistic approach, the framework supports student development, enhances learning outcomes, and prepares students for success in both academic and professional contexts.

Assessment tools

- This is an instrument that may be used by a teacher to determine the extent to which a leaner has achieved specified learning outcomes.
- The teacher may be guided by the following when choosing the Assessment tool to use;
  - *Knowledge and skills' evidence that the learner is expected to demonstrate*
  - *Performance criteria and the expected evidence*
  - *Assessment conditions*

In a Competence-Based Curriculum (CBC), assessment tools are designed to evaluate students' proficiency in specific competencies—skills, knowledge, and attitudes essential for success in real-world contexts. These tools help measure how well students can apply what they've learned in practical situations, providing a comprehensive view of their capabilities.

CBA Tools are instruments used to:

- Inform improvements on instruction strategies.
- Offer relevant and appropriate interventions
- Determine the level of acquired competencies
- Provide feedback to the learner and other stake holders
- Collect information on acquisition of competencies by a learner.

CBA requires the use of a variety of assessment tools to address the *different learning styles of the learners this include:*

Questionnaires

- A questionnaire is a specific set of written questions which aims to extract specific information from the chosen respondents.
- The questions and answers are designed in order to gather information about attitudes, skills, knowledge, preferences and factual information about the learners.
- Questions are either open-ended or closed.
- *Can be developed and administered by the teacher before, during or after the learning process*.

For effective use of Assessment questionnaire

- Ensure the assessment relates to the learning outcomes.
- Select the right type of question e.g. True/false, multiple-choice.
- Keep the questions simple, clear and concise.
- Provides the learner with useful feedback.
- Use scenario-based questions where possible that will make learners to apply what they've learned to realistic situations.

Sample Questionnaire

Administrative Information			
Learner's name:	Paula Pauline	Grade:	**7**
School:	Exel Junior Secondary School		
Learning Area/Subject:	Integrated Science		

Strand:	Force and Energy	**Sub-strand:**	Static electricity
Learning Activity:	Charging objects using rubbing and induction methods.		

Observation schedules

It outlines the characteristics and the behaviour that a learner shows during the performance of a specific task.

The teacher records observations made on the behaviour of the learner on the spot as outlined in the rubrics.

Sample observation Schedule

Administrative details.....

Criteria	
1. 2 3.	
Observations	
1. 2. 3.	
Teacher's comment:	
Student's signature:	**Date:**
Teacher's signature:	**Date:**

Written tests

- Tasks designed according to pre-determined criteria to measure competencies.
- Elicit evidence on the acquisition of learning outcomes and competencies.
- They take form of multiple choice, short structured and short essay questions
- They are accompanied by a scoring guide

SAMPLE WRITTEN TEST

The picture below was drawn from observation:

Identify two art elements used in the picture.....

2. After two minutes of play, Anita complained of severe pain in her ankle.

Write what could have caused the pain.....

Learner's profile

- A summary of the teacher's opinion on a learner's mastery of competencies.
- Enables the teacher to understand the:
  - i) competencies developed
  - ii) challenges the learner is experiencing
- The learner's competencies can be assessed by peers, teachers, parents, and community members.
- The teacher constructs a learner's profile using information obtained from:
  - observation schedule
  - learner's journal
  - checklist
  - portfolio
  - involvement in projects

Sample Learner Profile

Leaner's Name: Farida Njeri Naliaka **Grade: 7** **Teacher: Tr. Irungu Mwashetani** **Learning area/Subject:** **Strand:** **Sub-Strand:** **Learning outcome:**			
Criteria	**Students' strengths**	**Students' area of improvement**	**Students' Preferences/Interests**

Rating scales

- A rating scale is a tool used for assessing the performance of tasks, skill levels, procedures, processes, qualities, quantities, or end products, such as reports, drawings, and computer programs etc.
- These are judged at a defined level within a stated range.
- Just like a checklist, it can be used to assess performance or products, both formally and informally.
- The teacher states the **criteria** and provides **three** or **four** response selections to describe the quality or frequency of learner's work.
- The teacher then indicates the degree or frequency of occurrence of competencies displayed by the learner.
- Each scale is an incremental level of measurement, meaning, each scale fulfills the function of the previous scale.

Descriptive words for Rating scale

1**.**

- Always
- Usually
- Sometimes

2

- Excellent
- Good
- Fair

3.

- Outstanding
- Satisfactory
- Unsatisfactory

Meaning of some descriptive words used in rating scale

Level	**Rating**	**Description**
3	**Outstanding**	Performance exceeded the expectations and represents extra ordinary level of achievement and commitment, technical skills and knowledge, ingenuity, creativity and initiative. All goals, objectives and targets are achieved above the expected standards.
2	**Satisfactory**	Performance met expectations in terms of quality of work, efficiency and timelines. The most critical goals, objectives and targets are met.
1	**Unsatisfactory**	Performance was consistently below expectations and no reasonable progress was made towards achieving the critical goals and a significant improvement is needed in one or more important areas.

Sample Rating Scale

a) **Administrative Information**
Learner's name: *Yusuf Balaji* **Grade:** 7 **School:** *Exel Junior Secondary School* **Learning Area/Subject:** **Strand:** **Sub-strand:** **Learning Activity: A project on ch**

SKILL	CRITERIA	4 Always	3 Sometimes	2 Rarely	1 Never
Collaboration (individual score)	1. Works positively with others during the discussion/project activities.				
	2. Behaves appropriately during discussion/project activities				
	3. Is timely for the discussion and in carrying out the project activities				
	4. Is reliable/dependable to the group during the project activities				
	5. Completes assigned tasks during the discussion/project activities				
	6. Supports those who have difficulty during the project activities				
	7. Understands and respects diversity of others (culture, religion, beliefs)				
	8. Contributes to help the group make decisions				
	9. Recognizes the value and contribution of others during the discussion				
	10. Helps resolve conflicts during the discussion/project activities				
	11. Influences the group by tracking progress and suggesting improvements or corrective actions				

Comments on the learner's performance
Learner's signature _____ **Date** _____
Teacher's Name _____ **Signature**_____ **Date** _____

Oral or Aural questioning (Signing & Observing Assessment)

Oral/Signed Assessment

- Aimed at assessing a learner's speaking/signing and listening/observing skills.
- Teacher asks questions verbally/signing.
- Learner can respond verbally/signing or using other appropriate modes of communication.

- Teacher should give prompts and verbal/signed or non-verbal cues as they ask oral/signed questions

Aural/Observing Assessment

- Aimed at assessing the learners listening/observing skills.
- Usually pre-recorded (with captions) and played to the learner.

Strategies for effective Oral and Aural (Observing and Signing) Assessment

- Give a learner time to prepare and respond after the task is stated
- Ask a variety of questions:
  - Open-ended questions;
  - Questions that require more than a right or wrong answer;
  - Questions that promote higher-order thinking.

Administrative Information			
Learner's name:	*Fatma Hassan*	**Grade:**	
School:	*Al-Noor Junior Secondary School*		
Learning Area/Subject:	***Business Studies***		
Strand:	***Business and its environment***		
Learning Activity:	**Learners are given opportunity to visit any local market and compile a list of goods and services provided. The teacher then asks the learners the following questions after the visit. The leaners are expected to respond orally.** (i) What are some of the goods that you observed in the market? (ii) State some of the services that you observed during the visit? (iii) Can you now, distinguish between the terms "goods" and "services"? (iv) What are some of the goods and services that you did not observe but you would recommend to be provided in the market.		

ORAL ASSESSMENT

	Criteria	**OBSERVATION**
1	What are some of the goods that you observed in the market?	
2	State some of the services that you observed during the visit?	
3	Can you now, distinguish between the terms goods and services?	
4	What are some of the goods and services that you did not observe but you would recommend to be provided in the market.	

Teachers' comment on the learners performance:	
Teachers' signature:	
Students' signature:	

Anecdotal Records

- Anecdotal Records, provide teachers with a collection of narratives that can be used to showcase a child's progress over time.
- Describe the child's behavior NOT what you think of the behaviors. Use details of the child's behavior such as actions or comments. Write down the exact words used in the conversation
- The event may be described using Reports, photos, drawings etc.
- Describe the learner's behavior NOT what you think of the behaviors. Use details of the learner's behavior such as actions or comments. Write down the exact words used in the conversation where possible.
- **ANECDOTAL RECORDS**
- **School**:……..…
- **Learner's Name**:……**Grade**……
- **Observation Date**: ……. **Observation Time**: ……
-
- **Description of the incident/event**
- **……**
- **Description of location/Setting**
- **……**
- **……**
- **Teacher Comments:**
- **……**
- **Teacher's Name_____Signature_____**

SAMPLE ANECDOTAL RECORDS

School: **KIWA JUNIOR SECONDARY SCHOOL**

Learner's Name: **BWANA BWANA** **Grade**: **7**

Observation Date: **14/11/2022** **Observation Time**: **11.45 AM**

Description of the incident/event

Toby walked over to the sandpit and picked up a toy truck. Carrying the truck with both of his hands, Toby walked over to two of his friends who were also in the sandpit. Toby said "I got a dump truck, there is more over there, get one too" he said to the others pointing towards where the trucks were. Toby sat in the sandpit, put the truck in front of him and used his hands to fill the truck with sand. After filling the truck, he pushed it

across the sandpit while kneeling. “Too heavy, hard to push.” He commented. He stopped pushing and then begun to remove some of the sand from the truck. He then tried to push it across the sandpit again. “Look I can push it faster now.” Toby said.

Description of location/Setting

Playing in the sandpit with his friends

Teacher Comments:

Toby is able to explore his environment, manipulates objects and experiments. He openly explores feelings and ideas in his interaction with friends. He can initiate and joins in play. Persists when faced with challenges and when first attempts are not successful. He has interest in others and can be part of the group.

Teacher’s Name: _____ Signature: _____

Journals

- A journal entails the learner keeping a record of their personal feelings, thoughts, experiences and activities on a daily basis. *Records of past events or happenings in a learner’s life.*
- Learner’s development can be tracked through their writings.
- Based on the learner’s performance/feelings/thoughts, the teacher can provide either support, challenge or both.
- Assessment through journals should be a joint venture between the learner and the teacher

Sample Assessment Journal

School:	Sina Padua Junior Secondary School
Learner’s Name:	Chol Awol Babu
Grade	7
Date of Entry:	3/05/2022
Targeted Competency:	Speaking Skills
Happenings/Activities:	Public Speaking Competition
Learners feelings:	Today I felt embarrassed, because I was not able to present my work well.
Teacher’s comment to the learner	You are a talented learner, do more practice on public speaking.
Teacher’s recommendation	Awol should be given more opportunity to make presentations. This will help him develop courage to be able to speak before a large group of people.

Projects

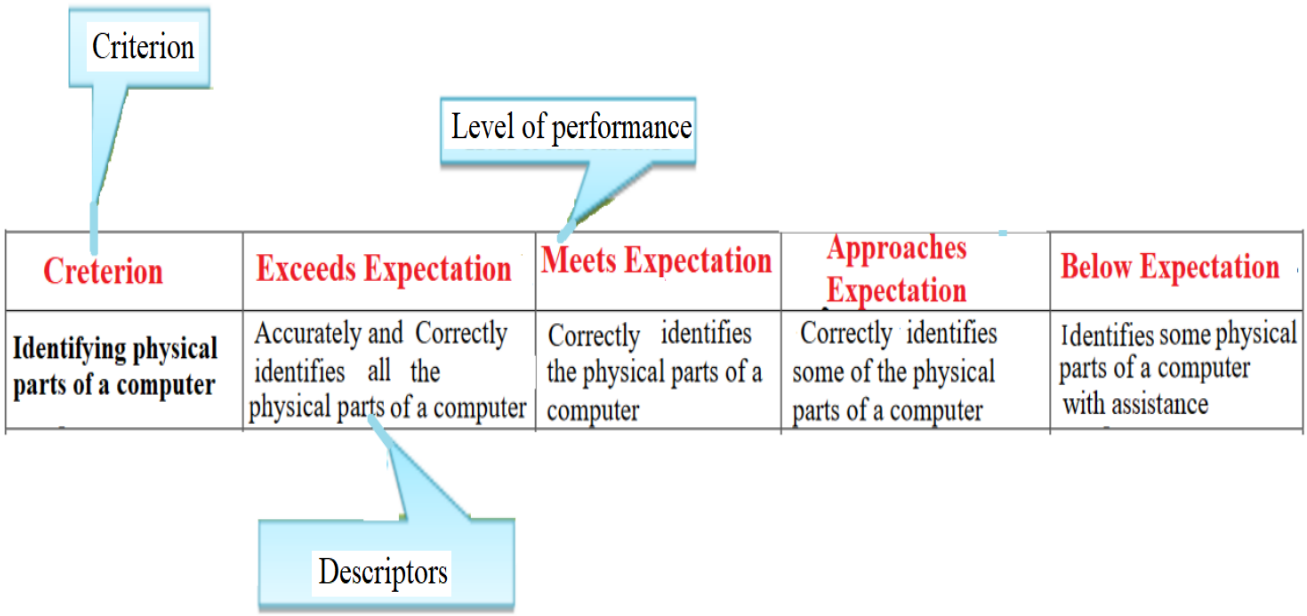
- A Project is a set of activities implemented within a given timeframe.
- Gives an opportunity for learners to apply acquired knowledge and skills to a real life situation.
- Should have a clearly stated purpose and set of objectives
- Can be assessed using observation schedules, rating scales, checklists, journals and portfolios
- Projects can be used for learning, problem solving or income generation

Administrative Information			
Learner's name:	**Kamaa Jamal**		
School:	**CPU Junior School**		
Subject:	**Computer Science**		
Strand:	**Foundation of Computer Science**		
Activity: Use a computer to type the document given below and print 2 copies. Submit your work after one week.			

ASSESSMENT RUBRIC

- It's an assessment tool used to interpret and assess learner's performance in a task. It consists of:

Criteria **Descriptors** **Performance Levels**



Creterion	Exceeds Expectation	Meets Expectation	Approaches Expectation	Below Expectation
Identifying physical parts of a computer	Accurately and Correctly identifies all the physical parts of a computer	Correctly identifies the physical parts of a computer	Correctly identifies some of the physical parts of a computer	Identifies some physical parts of a computer with assistance

DESCRIPTORS

Descriptors for high performance level

- Presents important details/facts/concepts
- Provides solution without errors
- Exhibits mmastery of knowledge and extensive understanding
- Uses multiple methods to solve a problem
- Represents a variety of perspectives
- Draws complex connections and logical conclusions which are not immediately obvious
- Provides clear and elaborate details with precision and accuracy
- Uses multiple sources to solves problems by effective application of information

Descriptors for low performance level

- Gives details with some errors
- Solves problems without complete understanding
- Generally explains
- Uses a single method
- Represents a single perspective
- Identifies few connects
- Without drawing accurate conclusions
- Presents confusing statements and facts
- limited and inaccuracies details
- Relies on single source

There are two types of rubrics.

1. **Analytic Rubric**
   - Breaks down content or tasks being assessed into smaller parts.
   - Each part is assessed separately.
2. **Holistic Rubric**
   - Assesses overall performance on a task as a single entity.
   - Scores the overall competencies of the learner.

Portfolio

- A portfolio is a purposeful collection of learner's work.
- Shows effort, progress and mastery of a specified learning outcomes or competencies.
- It provides evidence of achievement or mastery over a period of time.

Types of Portfolios

- **Working Portfolio:** A collection of learner's work in progress or completed samples of work
- **Showcase Portfolio:** Created by a learner to display their best work
- **Assessment Portfolio**: Used to document a learner's mastery of skills or competences in a particular learning area. It contains samples of learner's work that has been assessed by the teacher.

STAGES OF DEVELOPING A PORTFOLIO

1. **Planning**
   i. Type
   ii. Audience
   iii. Storage
   iv. Communication
2. **Collection of work samples**
3. **Selection of work samples**
4. **Reflection**
5. **Feedback**

Effective Management of Portfolios

The teacher should;

- Review entries in the portfolio with learners regularly.
- Focus more on quality than quantity.
- Involve the learners in organizing the portfolio.
- Where possible, organize for the storage of the learners' portfolios

Storage of Portfolios

Learners' portfolios can be organized and stored in:

- Ring binders
- Cartons
- Storage holders
- Media platforms
- Electronic storage devices
- Files/folders

Importance of Portfolio Assessment

- Enables learners to show quality of work in collaboration with others
- Instructional goals are set at the beginning and shared with the learner
- Provides a clear profile of learners in terms of abilities and interests
- It allows demonstration of a wide range of competencies
- It demonstrates leaner's progress over time.
- It gives learners an opportunity to reflect on their work.
- It enables learners to demonstrate their creativity

Checklists

- *Checklists* are used for identifying whether key tasks in a procedure, process, or activity have been completed.
- A teacher carefully observes and tick whatever behavior is portrayed guided by the criterion set.
- It is used to assess performance or products, both formally and informally.
- It offers a Yes/No, True/False format in relation to a learner demonstrating a specific criteria.

- Contains a list of characteristics of a learner's expected behavior
- Used to gauge skills development or progress.
- Requires the teacher's careful observation of the competences.
- Teacher marks/ticks against the competencies portrayed.

Characteristics of a good checklists

Should**:**

- Have criteria for success based on expected outcomes.
- Be short enough to be practical (e.g., one sheet of paper).
- Have tasks chunked into logical sections or flow from start to finish.
- Highlight critical tasks only.
- Be written with clear and detailed wording to minimize the risk of misinterpretation.

Sample Checklist

a) **Learning Activity/Task**

No.	Name	Competence (Knowledge, skills, attitude, values) assessed (tick appropriately)								
										Teacher's comment
		YES	NO	YES	NO	YES	NO	YES	NO	
1.										
2.										
	Teacher's signature							Date		

Implementing Assessment Tools in CBC

1. **Align with Competencies**:
   - Ensure that all assessment tools are aligned with the specific competencies outlined in the curriculum. This alignment ensures that assessments accurately measure the desired skills and knowledge.
2. **Provide Clear Criteria**:
   - Develop and communicate clear criteria and expectations for each assessment tool. This clarity helps students understand what is required and how their performance will be evaluated.
3. **Incorporate Diverse Methods**:
   - Use a variety of assessment tools to capture different aspects of competency. Combining different methods provides a more comprehensive view of students' abilities.

4. **Ensure Transparency**:
    - Make the assessment process transparent to students. Provide them with rubrics, checklists, and examples to guide their work and understanding of how they will be assessed.
5. **Encourage Reflection**:
    - Integrate opportunities for reflection in assessments. Encourage students to analyze their performance, set goals, and identify areas for improvement.
6. **Provide Constructive Feedback**:
    - Offer timely, specific, and actionable feedback on assessments. Feedback should help students understand their strengths, address weaknesses, and improve their performance.
7. **Regularly Review and Adjust**:
    - Regularly review the effectiveness of assessment tools and make adjustments as needed. Gather feedback from students and educators to refine assessment practices and ensure they meet the needs of the curriculum.

Benefits of Competence-Based Assessment Tools

1. **Real-World Relevance**:
    - Assessments are designed to reflect real-world tasks and challenges, ensuring that students develop practical and applicable skills.
2. **Comprehensive Evaluation**:
    - A range of assessment tools provides a holistic view of students' competencies, capturing different aspects of their performance.
3. **Student Engagement**:
    - Authentic and varied assessment methods increase student engagement and motivation by making learning more relevant and meaningful.
4. **Personalized Learning**:
    - Competence-based assessments allow for personalized feedback and support, helping students progress at their own pace and according to their individual needs.
5. **Preparation for Future Success**:
    - By focusing on competencies and real-world applications, assessment tools prepare students for success in their future careers and lives.

Challenges and Considerations

- **Consistency**: Ensuring consistency in the application and interpretation of assessment tools can be challenging. Clear guidelines and training for assessors are essential.
- **Resource Intensive**: Some assessment methods, such as performance-based assessments and projects, may require significant time and resources to implement effectively.
- **Complexity**: Designing and implementing a comprehensive competence-based assessment framework can be complex and may require ongoing adjustments and refinements.

Conclusion

Competence-Based Assessment Tools are crucial for evaluating students' proficiency in a Competence-Based Curriculum. By focusing on real-world applications, providing clear criteria, and using a variety of assessment methods, these tools offer a comprehensive and meaningful measure of students' competencies. Implementing these tools effectively ensures that assessments are aligned with curriculum goals, supports student learning, and prepares students for success in their future endeavors.

Topic 10

ICT integration in Education

Information and Communication Technology (ICT) integration in education within a Competence-Based Curriculum (CBC) enhances teaching and learning by leveraging digital tools and resources to support the development of specific competencies. ICT can transform how educators deliver content, how students engage with learning materials, and how assessments are conducted. Here's how ICT integration can be effectively implemented in a CBC:

Benefits of ICT Integration in CBC

1. **Enhanced Learning Experiences**:
    - **Interactive Content**: Digital tools such as multimedia presentations, interactive simulations, and educational games can make learning more engaging and interactive.
    - **Access to Resources**: Students have access to a wide range of online resources, including academic journals, educational videos, and interactive platforms.
2. **Personalized Learning**:
    - **Adaptive Learning Technologies**: Tools that adapt to individual students' learning needs and progress, offering tailored content and assessments based on their performance.
    - **Self-Paced Learning**: Online platforms allow students to learn at their own pace, providing opportunities for remediation or advanced study as needed.
3. **Collaboration and Communication**:

- **Online Collaboration Tools**: Platforms like Google Workspace, Microsoft Teams, and collaborative software enable students to work together on projects, communicate effectively, and share resources.
- **Virtual Classrooms**: Technologies such as video conferencing facilitate remote learning and collaboration, allowing for interaction with peers and instructors regardless of location.

4. **Efficient Assessment and Feedback**:
   - **Digital Assessments**: Tools for creating and administering quizzes, tests, and assignments online make the assessment process more efficient and flexible.
   - **Automated Feedback**: Online assessment tools can provide immediate feedback to students, helping them understand their performance and areas for improvement quickly.
5. **Real-World Skills Development**:
   - **Digital Literacy**: Using ICT tools helps students develop essential digital literacy skills, including information search, data management, and online communication.
   - **Technical Skills**: Exposure to various digital tools and platforms prepares students for future careers in a technology-driven world.

Strategies for Integrating ICT in CBC

1. **Align ICT with Competencies**:
   - **Curriculum Integration**: Ensure that ICT tools and resources are aligned with the competencies outlined in the CBC. This alignment ensures that technology supports the development of relevant skills and knowledge.
   - **Learning Objectives**: Design ICT activities that align with specific learning objectives and competencies, providing opportunities for students to apply their skills in meaningful ways.
2. **Utilize Digital Tools and Resources**:
   - **Educational Software**: Incorporate software and applications designed for educational purposes, such as interactive simulations, virtual labs, and language learning tools.
   - **Online Resources**: Use online databases, digital libraries, and educational websites to provide students with access to a wide range of information and learning materials.
3. **Facilitate Collaborative Learning**:
   - **Group Projects**: Encourage students to use online collaboration tools for group projects and assignments, promoting teamwork and communication skills.
   - **Discussion Forums**: Create online discussion forums and groups where students can engage in discussions, share ideas, and collaborate on problem-solving activities.
4. **Implement Flipped Classroom Models**:

- **Pre-Learning Materials**: Use online videos, readings, and interactive content as pre-learning materials that students review before class, allowing for more interactive and application-focused classroom activities.
- **In-Class Activities**: Use class time for hands-on activities, discussions, and problem-solving exercises, leveraging ICT tools to support and enhance these activities.

5. **Develop Digital Assessments**:
    - **Online Quizzes and Tests**: Create and administer quizzes, tests, and assignments online to assess students' understanding and application of competencies.
    - **E-Portfolios**: Implement digital portfolios where students can collect and showcase their work, reflecting on their learning and demonstrating their competencies.
6. **Provide Professional Development for Educators**:
    - **Training Programs**: Offer professional development opportunities for educators to learn how to effectively integrate ICT into their teaching practices and curriculum.
    - **Ongoing Support**: Provide ongoing support and resources to help educators stay updated with the latest technologies and best practices in ICT integration.
7. **Ensure Accessibility and Inclusivity**:
    - **Accessible Technologies**: Use ICT tools and resources that are accessible to all students, including those with disabilities. Ensure that digital content is compatible with assistive technologies.
    - **Inclusive Practices**: Design ICT activities and assessments that accommodate diverse learning styles and needs, providing equitable opportunities for all students.
8. **Monitor and Evaluate ICT Integration**:
    - **Feedback and Evaluation**: Regularly gather feedback from students and educators on the effectiveness of ICT integration. Evaluate how well digital tools and resources support the achievement of competencies.
    - **Adjustments and Improvements**: Make necessary adjustments based on feedback and evaluation results to improve the integration of ICT in the curriculum.

Examples of ICT Integration in CBC

1. **Virtual Labs**:
    - Science students use virtual lab simulations to conduct experiments, allowing them to explore concepts and perform tasks that may be impractical in a physical lab setting.
2. **Interactive Whiteboards**:
    - Teachers use interactive whiteboards to present lessons, demonstrate concepts, and engage students in interactive activities.
3. **Educational Apps**:

    - Language learners use apps for vocabulary practice, grammar exercises, and language games, supporting personalized learning and skill development.
4. **Online Collaboration Platforms**:
    - Students collaborate on research projects using platforms like Google Docs or Microsoft Teams, sharing documents, and working together in real time.
5. **E-Learning Modules**:
    - Students complete e-learning modules on topics related to their curriculum, accessing multimedia content and interactive exercises to enhance their understanding.

Challenges and Considerations

1. **Technology Access and Equity**:
    - Ensuring that all students have access to the necessary technology and internet connectivity can be a challenge. Schools should address equity issues and provide support for students who may lack access.
2. **Training and Support**:
    - Adequate training and ongoing support for educators are crucial to the successful integration of ICT. Without proper training, educators may struggle to effectively use digital tools and resources.
3. **Privacy and Security**:
    - Protecting students' privacy and ensuring the security of digital tools and platforms is essential. Schools should implement policies and practices to safeguard sensitive information.

4. **Balancing Screen Time**:
    - Managing the amount of screen time to ensure it is balanced with other forms of learning and physical activity is important for students' overall well-being.

Conclusion

Integrating ICT in a Competence-Based Curriculum enhances teaching and learning by providing interactive, personalized, and efficient educational experiences. By aligning digital tools with competencies, facilitating collaboration, and providing effective training and support, educators can leverage ICT to improve student outcomes and prepare them for success in a technology-driven world. Despite challenges, thoughtful and strategic ICT integration can significantly enrich the learning process and help students develop the competencies needed for their future careers and lives.

Topic 11

Introduction to Comprehensive School Health and safety

Comprehensive School Health and Safety involves a holistic approach to ensuring the well-being and safety of students, staff, and the school environment. This approach integrates various aspects of health, safety, and wellness into the school's operations and culture, aiming to create a supportive and secure environment conducive to learning and development.

Key Components of Comprehensive School Health and Safety

1. **Physical Health**:
    - **Nutrition**: Ensuring that students have access to healthy food options, including nutritious meals and snacks, and promoting healthy eating habits.
    - **Physical Activity**: Providing opportunities for regular physical exercise through physical education classes, sports, and recreational activities.
    - **Health Services**: Offering on-site health services, including school nurses and health clinics, to address medical needs and provide health education.
2. **Mental and Emotional Health**:
    - **Counseling Services**: Providing access to school counselors and mental health professionals to support students' emotional well-being and address issues such as stress, anxiety, and depression.
    - **Social-Emotional Learning (SEL)**: Integrating SEL programs into the curriculum to help students develop skills for managing emotions, building relationships, and making responsible decisions.
3. **Safety and Security**:
    - **Emergency Preparedness**: Developing and implementing emergency response plans for various scenarios, including natural disasters, medical emergencies, and security threats.
    - **Building Safety**: Ensuring that school facilities are safe, well-maintained, and compliant with building codes and safety regulations. This includes regular inspections and maintenance of infrastructure and equipment.
    - **Cybersecurity**: Protecting students' and staff's digital information and ensuring safe use of technology through policies and practices that address online safety and data security.
4. **Health Education**:
    - **Curriculum Integration**: Incorporating health education into the curriculum to teach students about topics such as nutrition, personal hygiene, mental health, substance abuse prevention, and healthy lifestyles.
    - **Parental Involvement**: Engaging parents and guardians in health education initiatives and providing resources to support healthy behaviors at home.
5. **Safe and Supportive Environment**:
    - **Anti-Bullying Policies**: Implementing and enforcing policies and programs to prevent and address bullying and harassment, ensuring a safe and inclusive school environment for all students.

- **Positive Behavior Support**: Promoting positive behavior and creating a supportive school culture through programs and strategies that encourage respect, responsibility, and kindness.

6. **Involvement and Collaboration**:
    - **Stakeholder Engagement**: Involving students, parents, staff, and community members in developing and implementing health and safety policies and programs. Collaboration with local health agencies and organizations can also enhance school health initiatives.
    - **Professional Development**: Providing ongoing training and professional development for staff on health and safety topics, including first aid, mental health support, and emergency response.

Benefits of Comprehensive School Health and Safety

1. **Improved Academic Performance**:
    - A healthy and safe school environment supports students' ability to focus on learning, leading to better academic outcomes and overall school performance.
2. **Enhanced Well-Being**:
    - Addressing physical, mental, and emotional health needs helps students feel more secure, supported, and engaged in their education.
3. **Reduced Absenteeism**:
    - Health and safety initiatives can reduce the incidence of illness and injury, leading to fewer missed school days and improved attendance rates.
4. **Stronger School Community**:
    - A comprehensive approach fosters a positive and supportive school culture, enhancing relationships among students, staff, and families.
5. **Preparedness for Emergencies**:
    - Effective emergency preparedness plans and practices ensure that the school is ready to respond to crises, minimizing potential risks and impacts.

Implementing a Comprehensive School Health and Safety Program

1. **Assessment and Planning**:
    - Conduct a thorough assessment of current health and safety practices and identify areas for improvement. Develop a comprehensive plan that outlines goals, strategies, and responsibilities for implementing health and safety initiatives.
2. **Policy Development**:
    - Establish clear policies and procedures related to health and safety, including emergency response, health education, and anti-bullying measures. Ensure that policies are regularly reviewed and updated.
3. **Resource Allocation**:
    - Allocate resources, including funding, staff, and materials, to support health and safety programs. Invest in infrastructure improvements, training, and health services as needed.
4. **Monitoring and Evaluation**:

    - Regularly monitor and evaluate the effectiveness of health and safety programs. Gather feedback from students, staff, and parents to identify successes and areas for improvement.
5. **Communication and Awareness**:
    - Communicate health and safety policies and initiatives to the school community. Use various channels, such as newsletters, meetings, and workshops, to raise awareness and promote engagement.

Challenges and Considerations

1. **Funding and Resources**:
    - Securing adequate funding and resources for health and safety programs can be challenging. Schools may need to explore funding opportunities, partnerships, and community support.
2. **Compliance and Regulation**:
    - Ensuring compliance with health and safety regulations and standards requires ongoing attention and effort. Schools must stay informed about legal requirements and best practices.
3. **Cultural and Socioeconomic Factors**:
    - Addressing the diverse needs of students from different cultural and socioeconomic backgrounds may require tailored approaches and additional support.
4. **Staff Training**:
    - Providing effective training for staff on health and safety topics is essential but can be resource-intensive. Ongoing professional development and support are necessary to maintain staff competency.

Conclusion

Comprehensive School Health and Safety is a vital aspect of creating a positive and supportive learning environment. By addressing physical, mental, and emotional health needs, ensuring safety and security, and promoting health education, schools can enhance students' well-being and academic success. Effective implementation involves careful planning, collaboration, and ongoing evaluation to ensure that health and safety initiatives meet the needs of the school community and contribute to a thriving educational environment.

Topic 12

Preparation of professional documents

In the context of a Competence-Based Curriculum (CBC), professional documents play a crucial role in guiding and documenting educational practices, assessing student progress, and ensuring alignment with curriculum objectives. These documents are essential for effective curriculum delivery, assessment, and reporting. Here's an overview of key professional documents and how to prepare them in a CBC framework:

Key Professional Documents in CBC

1. **Curriculum designs**
    - **Purpose**: Provide an overview of the curriculum, including learning outcomes, content, teaching strategies, and assessment methods.
    - **Preparation**:
        - **Align with Competencies**: Ensure that the curriculum guide aligns with the specific competencies outlined in the CBC. Clearly define the competencies to be achieved at each grade level or course.
        - **Include Learning Outcomes**: Specify what students are expected to know, do, and value by the end of the course or program.
        - **Detail Instructional Strategies**: Describe the teaching methods and resources to be used, including any ICT tools or differentiated instruction strategies.
        - **Outline Assessment Methods**: Specify how students' competencies will be assessed, including formative and summative assessments.
2. **Schemes of Work**
    - What document should guide the teacher when developing Schemes of Work?
    - Why should we prepare Schemes of Work?
    - Should Schemes of Work be written or handwritten?
    - Outline the components of a Schemes of Work.
    - What are the gaps in the preparation and use of schemes of work?

- Schemes of work are part and parcel of the teaching profession.
- The development of SOW and other professional documents is guided by the curriculum designs.
- This helps the teachers in their interpretation of the designs and subsequent implementation of the curriculum.
- They lay the foundation for sound administration, planning, monitoring and evaluation of curriculum delivery.
- Ideally, schemes of work should be prepared annually. Reason: They provide a roadmap of what the teacher will achieve over a long period of time.
- Schemes of work can either be written or handwritten. However, they should be updated on a daily basis.

Gaps

- Overreliance on online/commercial schemes of work. These do not address the needs and situation of the learner.
- Failure to use recommended format.
- Not aligning them to the curriculum design. Instead many teachers indicate that they use course books for scheming purposes.
- Failure to quote the exact reference being used including title, page numbers, author, when using course books and other reference materials
- Stating lesson specific learning outcome using sub strand instead of lesson.
- Schemes of work are done to comply with documentation hence not personalized.
- Reflection column is normally not updated. If so, it is sketchy with remarks such as done, covered which cannot inform teaching and improvement.
- Failure to update schemes of work over a period of time.
- There is disregard of the three levels of the specific learning outcomes. Teachers tend to focus on mainly knowledge.
- Some specific learning outcomes lack the context.
- Learning experiences in the scheme of work are **not** suggested since it is the teacher who has developed them to suit their contexts. But in the curriculum design they are suggested.

SCHEMES OF WORK TEMPLATE

Administrative Details

School	
Year	
Term	
Grade	

WK	**Lesson**	**Strand**	**Sub Strand**	**Specific Learning Outcomes**	**Learning Experiences**	**KIQ**	**Learning Resources**	**Assessment Methods**	**Reflection**

3. **Lesson Plans**

- A lesson plan helps to actualise what is visualised in the schemes of work. It ensures effective teaching and learning because it allows the teacher to pre-meditate on the content delivery beforehand.
- Preparing a lesson gives the teacher an opportunity to think through how they will incorporate critical elements of CBC such as *core competences, PCIs, values, non-formal learning activities, link to other learning areas, resources and assessment.*

Importance of lesson planning

A well-done lesson plan helps the teacher to:

- organize systematically the content to be taught in advance hence avoiding vagueness and irrelevance,
- plan, prepare and assemble teaching and learning resources,
- present concepts and skills in a systematic manner using appropriate strategies to achieve the stated lesson outcomes,
- manage time well during the lesson,
- select and design appropriate assessment methods to evaluate the teaching and learning process,
- To make connections between various components.

SAMPLE LESSON PLAN TEMPLATE

Administrative Details

School		**Date**	
Subject		Time	
Year		Grade	
Term		Roll	

Strand: **Sub Strand:** **Specific Learning Outcomes:** **Key Inquiry Questions:** **Learning Resources:** **Organisation of Learning:** **Introduction:** **Lesson Development:** **Step 1** **Step 2 {Mainstream Core Competencies, Values and PCIs}** **Step 3** **Step 4……** **Extended Activities:** **Conclusion:** **Reflection on the lesson:**

POINTS TO LOOK FOR IN A LESSON PRESENTATION

1. Are the Learning outcomes well unpacked from broader sub strand to bring out the intended knowledge, skill(s), attitudes and values?
2. Does the lesson plan have real life experiences well spelt out?
3. Are the core competencies(s) to be developed well-articulated and in a precise manner?
4. Are the Value(s) to be nurtured well-articulated and in a precise manner?
5. Are PCI'(s) to be mainstreamed well-articulated and in a precise manner?
6. What strategies or pedagogies will the teacher employ to deliver the lesson
7. How does the teacher assess the KSA?(Assessment should be integrated during the lesson e.g asking questions, responding to questions, writing answers, taking notes, summarising)
8. Are all the components of a lesson plan well brought out and in the correct format(e.g KIQ's, learning resources Lesson organisation etc)
9. How does the teacher keep records of the individual learner performance?

`

3. **RECORD OF WORK**

Provides evidence of work covered by the teacher, it should be updated on a daily basis.

Importance of record of work

- Provides evidence of work covered by the teacher thus ensuring **accountability and transparency.**
- Provides **continuity** of teaching of a particular class since a new teacher is able to trace where to start teaching.
- Helps identify the learners **who needs more help, guidance and support (** in the reflection column)

RECORD OF WORK COVERED TEMPLATE

Administrative Details

School	
Grade	
Subject	
Year	
Term	
Name of Teacher	

Date	**Strand**	**Sub Strand**	**Lesson**	**Work done**	**Reflection**	**Signature**

Individualised Education Programme (IEP)

Individualised Education Programme (IEP) is a written educational plan designed to meet the individual learning needs of a learner. It is a comprehensive and legally mandated document that outlines the specialised education plan for a learner with special need.

Rationale for developing an IEP

- IEP helps identify appropriate learning outcomes, experiences, resources and assessment to ensure that learners with special needs and disabilities are successful in learning.
- It guides the educational planning and monitoring of a learner's unique learning needs.
- It provides an opportunity to share information between the teacher, parent or caregiver and other professionals on how to improve educational results for learners with special needs.

Learners who may require an IEP include;

- Learners with learning disabilities (language, arithmetic, writing, reading, executive functioning)
- Learners with Attention Deficit Hyperactive Disorder (ADHD)
- Learner with Emotional and Behaviour Disorders (EBD)
- Learners with loss of vision or hearing senses

Structure of an IEP

1. Title: This is the name of the professional document "Individualised Education Programme"

2. Administrative details: This refers to details that give reference to the IEP

- Name of the School.
- Level or Grade.
- Learning Area or Subject:

3. Target Learners: This entails the name or names of learners targeted by the IEP.

4. Strand and Sub strand:

A strand is a broad area of study within a specific learning area or subject.

A sub strand represents smaller concepts within the strand. In the IEP state clearly the strand and sub- strand from the curriculum design.

5. Areas of strength and areas of need;

- Establish the strengths and areas of need for every learner using teacher made assessment tools such as observation schedules, checklists, questionnaires and rubrics among others.
- State them in terms of what the learner can do and areas of need that require intervention with reference to the curriculum designs.

6. Specific Learning Outcomes:

- This indicates the expected observable changes in the learner after covering a sub strand or a lesson.
- Based on the learning needs identified, develop specific learning outcomes for every learner in the target learning area or subject with reference to the curriculum design.
- The specific learning outcomes should cover the duration of the IEP depending on the ability of the individual learner.

7. Learning Experiences: This refers to activities that the learner is engaged in during the learning process to achieve the intended specific learning outcomes and develop competencies.

8. Resources: These are materials and devices that the learner interacts with in the learning process in order to achieve the desired outcomes and develop competencies. The teacher should indicate resources such as charts, realia, models and assistive technologies among others.

9. Assessment: These are the methods and tools used to determine the level of achievement of the learner as per the targeted specific learning outcome indicated in the IEP.

10. Date of initiation: This is the date when the implementation of the IEP begins.

11. Date of termination: This is the date when the period of the IEP ends.

12. Reflection: This indicates the teacher's comments regarding the following:

- ✓ Areas of success.
- ✓ Areas of difficulty.
- ✓ Intervention measures to be put in place to alleviate the difficulties.

STRUCTURE OF AN IEP

Administrative Details

Name of the School: _____ Grade or level: ___

Subject: ____

Name of the Learner	**Strand and sub strand from the designs**	**Areas of Strength**	**Areas of Need**	**Specific Learning Outcomes**	**Learning Experiences**	**Resources**	**Assessment (methods and tools)**	**Date of Initiation**	**Date of Termination**	**Reflection**
James										
Peter										

Mary										
Caroline										

Best Practices for Preparing Professional Documents

1. **Align with Competencies**:
   - Ensure that all documents are aligned with the competencies and learning outcomes outlined in the CBC. This alignment ensures that the documents support the achievement of curriculum goals.
2. **Be Clear and Specific**:
   - Provide clear, detailed information in all documents. Use specific language to define objectives, criteria, and expectations, making it easier for educators and students to understand and implement.
3. **Incorporate Feedback**:
   - Seek feedback from stakeholders, including students, parents, and colleagues, to refine and improve professional documents. Regularly review and update documents based on feedback and changing needs.
4. **Use Technology**:
   - Leverage digital tools and platforms to create, organize, and manage professional documents. This can enhance efficiency, accessibility, and collaboration.
5. **Ensure Consistency**:
   - Maintain consistency in the format and content of professional documents to ensure clarity and coherence across different documents and practices.
6. **Provide Training and Support**:
   - Offer training and support for educators on how to effectively use and implement the professional documents. This includes guidance on interpreting and applying the content of the documents.

Conclusion

The preparation of professional documents in a Competence-Based Curriculum (CBC) is essential for effective curriculum implementation, assessment, and reporting. By developing clear, detailed, and aligned documents such as curriculum guides, lesson plans, assessment tools, and portfolios, educators can support students' development of competencies and enhance the overall educational experience. Adopting best practices and regularly reviewing and updating documents ensure that they remain relevant and effective in achieving curriculum goals.

References

Competency-Based Education: A Guide for Educators by Rose Asera.

"What is Competence-Based Curriculum?" by UNESCO, which provides an overview of CBC principles and implementation strategies.

"Competence-Based Curriculum: Relevance, Implementation, and Challenges" available on platforms like Research Gate.

Designing and Implementing Competency-Based Education Programs by the Competency-Based Education Network (C-BEN).

UNESCO's *Competency-Based Education Toolkit* which offers guidelines and templates for CBC curriculum design.

Assessment for Competency-Based Education: Measuring Mastery by the American Institutes for Research.

Competency-Based Assessment in Higher Education by the Council for Adult and Experiential Learning (CAEL).

"Challenges in Implementing Competence-Based Curriculum: Perspectives from Educators" published in the *Journal of Education and Practice*.

UNESCO's case studies on the implementation of CBC in various countries, which highlight challenges and solutions.